Chapter 4

Dispersion analysis

One of the main tasks in surface wave testing is to extract information about the geometric dispersion of surface waves from observations of the particle motion at two or more receiver locations. The information is usually presented in the form of dispersion curves that show the variation of surface wave phase or group velocity versus frequency or wavelength. In this chapter, we describe the most widely used methods for calculating phase velocity dispersion curves. In fact, group velocities are rarely used for near-surface applications.

The methods are presented with reference to the analysis of Rayleigh waves, which are commonly analyzed for site characterization, but they may be easily extended to the analysis of other dispersive waves, such as Love waves, Scholte waves, and guided waves (see Chapter 8).

The methods may be classified according to the procedure adopted:

- Direct assessments of the propagation parameters (e.g., wavelength in the steady-state Rayleigh method or phase delay in the two-station spectral analysis of surface waves [SASW])
- Regression methods, in which the propagation parameters are obtained by fitting the experimental data with the expected theoretical functions (multi-offset phase analysis [MOPA], spatial autocorrelation [SPAC], transfer function)
- Transform-based methods, in which the experimental data are transformed from the original space–time domain into a different domain in which the propagation parameters are easily identified as spectral maxima (e.g., frequency–wavenumber, frequency–slowness, frequency–velocity analysis)

In this chapter, most of the examples are generated from a reference real dataset. The active-source multichannel test has been performed in Florence (Italy), using 24 vertical geophones, with a natural frequency of 4.5 Hz, deployed on a linear array with 2 m spacing. The wave field was generated with a weight-drop system, with a mass of 130 kg in free fall from 3 m. The seismic source had an offset of 4 m from the receiver array. 12 single shot

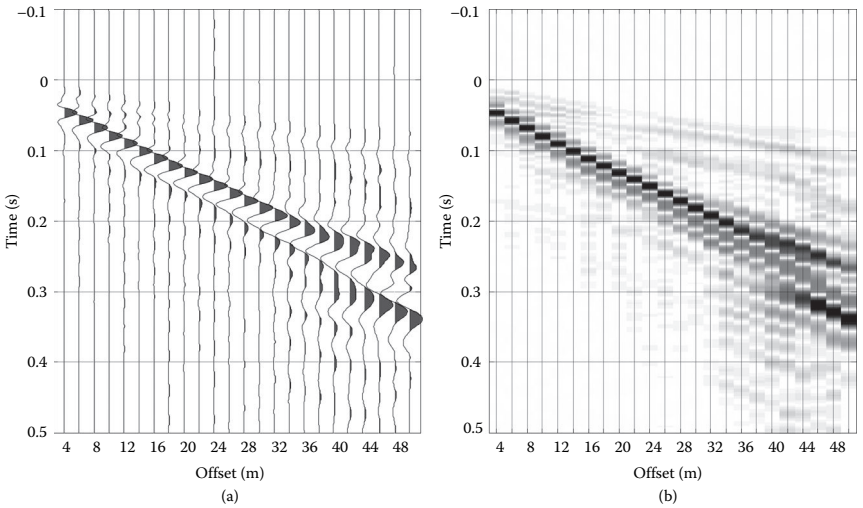

Figure 4.1 Reference dataset: stacked multichannel seismogram (shot gather) displayed as (a) wiggle traces and (b) density plot. The main surface wave event is the one arriving at the furthest offset at 0.3 s.

gathers have been acquired separately to allow for statistical assessment of signal-to-noise ratio (used in the SASW procedure, see Section 4.3) or for uncertainty evaluation (see Sections 4.4 and 4.8). The stacked shot gather is plotted in Figure 4.1.

4.1 PHASE AND GROUP VELOCITY

Nondispersive waves propagate with a single velocity, and their waveform propagates without changing shape. When there is dispersion, as for surface waves in heterogeneous media, the velocity depends on frequency, and the waveform changes shape as it propagates (see Chapter 2). Dispersive waves are characterized by two distinct velocities—the phase and group velocities (Rayleigh 1877). The phase velocity is the speed of propagation of a single phase of the waveform (e.g., a peak or trough), whereas the group velocity is the velocity of a packet or "group" of waves (Figure 4.2).

The phase and group velocity can be expressed mathematically as

$$V_{\text{phase}} = \frac{\omega}{k}$$
$$V_{\text{group}} = \frac{\partial\omega}{\partial k} \tag{4.1}$$

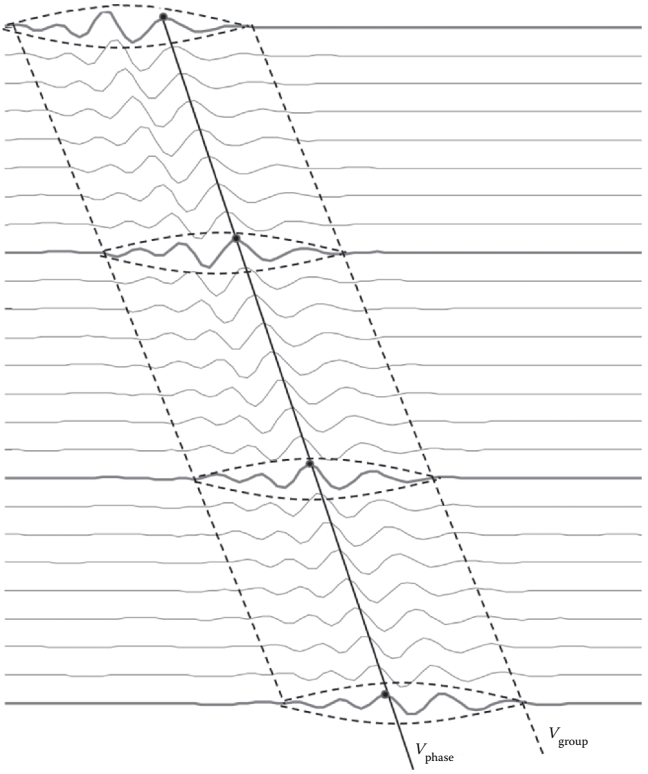

Figure 4.2 Phase velocity and group velocity.

The relationship between phase and group velocity is given by

$$\begin{aligned} V_{\text{group}} &= V_{\text{phase}} + k\frac{dV_{\text{phase}}}{dk} \\ &= V_{\text{phase}}\left(1 - k\frac{dV_{\text{phase}}}{d\omega}\right)^{-1} \end{aligned} \tag{4.2}$$

From Equation 4.2, it is clear that when the derivatives of phase velocity are equal to zero (i.e., the material is nondispersive), the phase and group velocities are identical.

For multimode surface waves, each mode can be described by its frequency-dependent phase and group velocities (i.e., phase and group dispersion curves). Figure 4.3 shows an example for two modes.

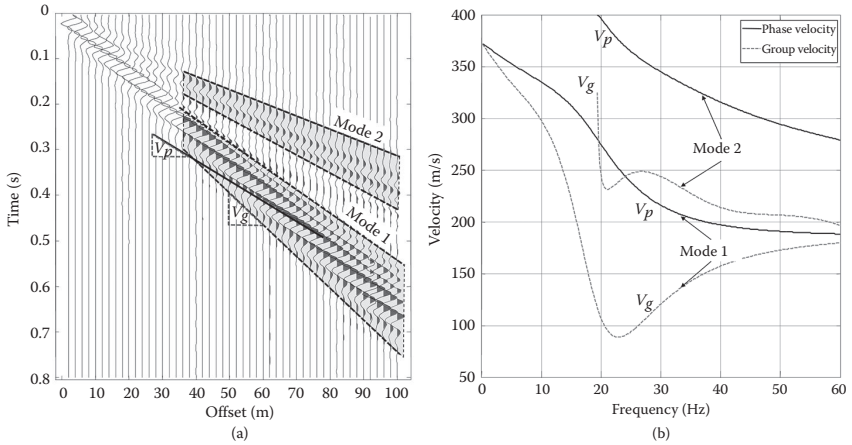

Figure 4.3 Example with synthetic data of surface waves with two dispersive modes: (a) synthetic shot gather; (b) phase velocity and group velocity dispersion curves.

From Equation 4.2, it is evident that the group velocity is lower than the phase velocity when the latter decreases with increasing frequency (as in the example of Figure 4.3). In this case, the system is defined as normally dispersive.

In the frequency–wavenumber domain, the position of a point corresponds to its phase velocity, while the tangent (i.e., the local slope) represents its group velocity. The example reported in Figure 4.4 shows that the points of the phase velocity dispersion curves with the same velocity (P1 to P4 in Figure 4.4a) are aligned along a straight line in the *f–k* domain (Figure 4.4b). On the contrary, the same value of group velocity for different modes (*G*1 to *G*3 in Figure 4.4a) corresponds to the same local derivative of the phase velocity dispersion curves in the *f–k* panel (Figure 4.4b). The *f–k* graph depicted in Figure 4.4b also shows constant phase velocity lines.

4.2 STEADY-STATE METHOD

The first surface wave method developed for near-surface site characterization was the steady-state Rayleigh method (SSRM), proposed by Jones (1958, 1962) and then adopted at the Waterways Experiment Station, in the United States (Ballard 1964).

Taking into account the limitation of available instrumentation at that time, the method was quite ingenious and allowed for an evaluation of the dispersion curve with a very simple experimental setup. Showing the potential of surface waves for site characterization, it paved the way for subsequent and more refined approaches that lead to the diffusion of surface waves in the engineering community.

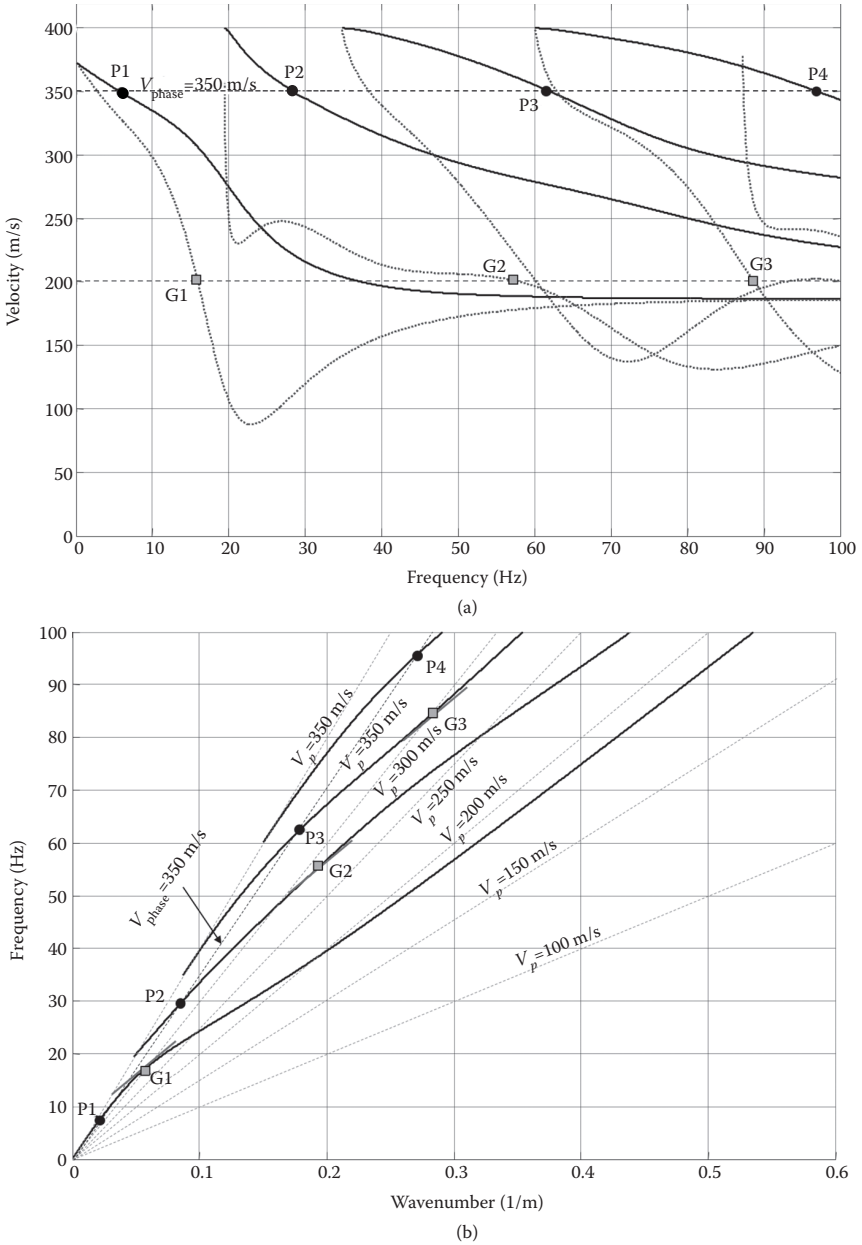

Figure 4.4 Example of surface waves with four dispersive modes. Phase velocity and group velocity dispersion curves are represented as (a) frequency versus velocity and (b) frequency versus wavenumber. Dotted straight lines in (b) represent constant phase velocity events.

In his field experiments, Jones used Rayleigh and Love waves to characterize the subsoil. In particular, the field equipment was composed of a mechanical vibrator and a single receiver. To investigate Rayleigh wave propagation, the vibrator was placed vertically so that the transmitted action was perpendicular to the free surface; for Love waves, the vibrator and the receiver were orientated to produce and detect vibrations in a horizontal direction transverse to the testing line. The source and the receiver were monitored through an oscilloscope.

The procedure of SSRM is illustrated in Figure 4.5. A vibrator, controlled by a sinusoidal input at a given frequency, is placed on the ground surface, and one receiver is used to detect the particle motion. The receiver is moved away from the vibrator until they are in phase. The distance between any two adjacent locations showing such a feature is assumed to be the wavelength at that particular frequency. Phase velocity is then evaluated as the product of wavelength and frequency. The procedure is then repeated and the operating frequency of the source is changed in order to reconstruct the experimental dispersion curve.

A robust estimate may be obtained considering several locations at which source and receiver are in phase (Figure 4.6). The slope of the straight line connecting the points in the diagram source-to-receiver distance versus the number of cycles represents the inverse of wavelength for the input frequency (Richart et al. 1970).

A similar approach, but with multiple receivers at fixed positions, is implemented in the continuous surface wave (CSW) method (Matthews et al. 1996; Menzies and Matthews 1996). In their procedure, the phase

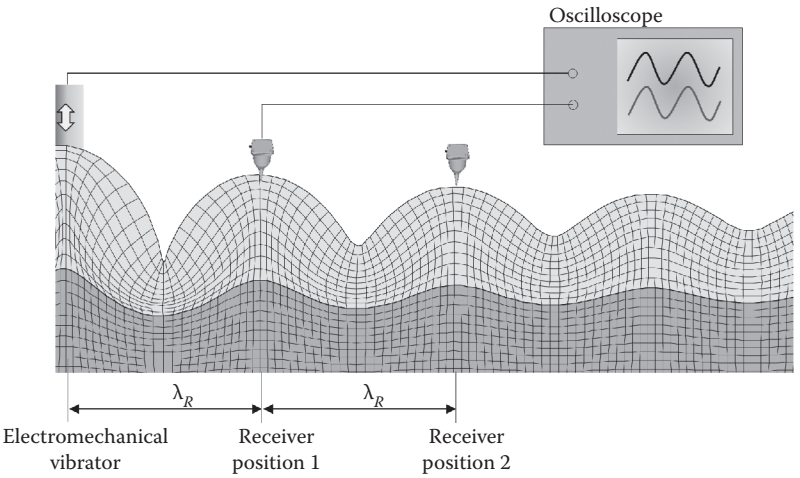

Figure 4.5 Steady-state Rayleigh method (SSRM), field procedure.

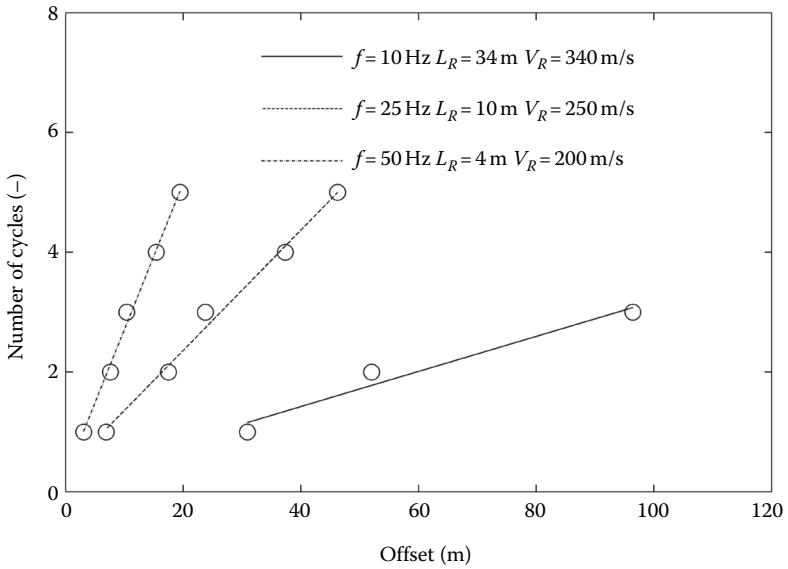

Figure 4.6 Determination of the average wavelength of Rayleigh waves by SSRM.

velocity is computed from the slope of the resulting phase angles plotted against source-to-receiver distance (see also Section 4.4).

4.3 SPECTRAL ANALYSIS OF SURFACE WAVES

Although the acronym SASW could in principle be used to designate any of the methods reported in the present chapter (because they all are based on the analysis of the propagation of surface waves in the frequency domain), it is conventionally used to address the following two-station procedure, which was proposed by researchers at the University of Texas, Austin, in the 1980s (Nazarian and Stokoe 1984; Stokoe et al. 1994).

SASW tests are performed with two receivers that are colinear with an active source (as shown in Figure 4.7). Typically, the receivers are placed such that the source offset (D) is equal to inter-receiver spacing (X) (Sànchez-Salinero 1987). The basic principle is that the velocity can be estimated as the ratio of the distance divided by time delay. For a single harmonic wave, the time delay is evaluated on a given phase of the signal. The method can be implemented using either harmonic sources or impulsive sources. The advantage of using harmonic sources is that the energy is concentrated at a given frequency, allowing high reliability in the determination. The advantage of using an impulsive source is that information over a wide range of

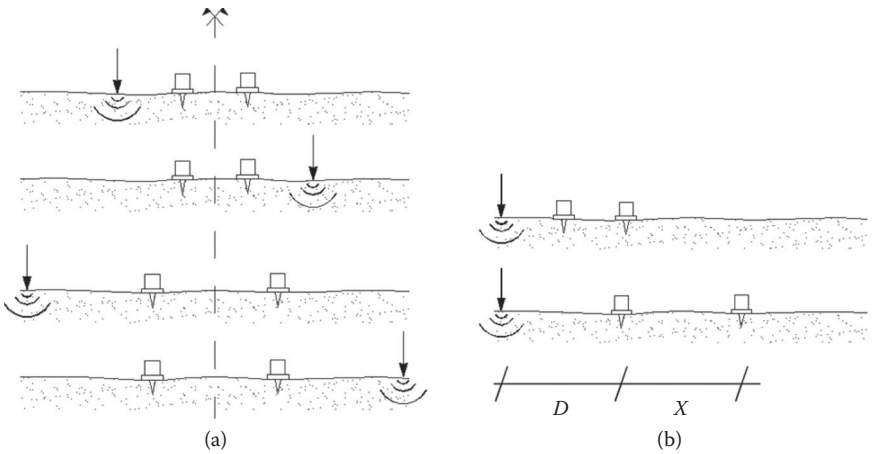

Figure 4.7 Acquisition schemes for two-station SASW tests: (a) common receivers midpoint; (b) common source.

frequencies is obtained with a single acquisition, allowing fast operations in the field.

The time domain signal recorded at each receiver is given by $s(x_m, t)\, m = 1,2$. Taking the temporal Fourier transform of each signal yields

$$S(x_m, \omega) = |S(x_m, \omega)| e^{i[\phi(\omega) + k(\omega) x_m]} \tag{4.3}$$

where $\phi(\omega)$ is the arbitrary phase of the source. The cross-power spectrum between the two signals is

$$\begin{aligned} S_{12}(\omega) &= \bar{S}_1(\omega) S_2(\omega) \\ &= |S(x_1, \omega)| e^{-i[\phi(\omega) + k(\omega) x_1]} |S(x_2, \omega)| e^{i[\phi(\omega) + k(\omega) x_2]} \\ &= |S(x_1, \omega)| |S(x_2, \omega)| e^{ik(\omega)(x_2 - x_1)} \end{aligned} \tag{4.4}$$

where $\bar{S}$ denotes the complex conjugate of S. In practice, it is desirable to use ensemble averaging to reduce the variance of the measured cross-power spectra (Bendat and Piersol 2010)

$$\hat{S}_{12}(\omega) = \frac{1}{N} \sum_{k=1}^{N} \bar{S}_k(x_1, \omega) S_k(x_2, \omega) \tag{4.5}$$

where N is the number of spectra that are averaged. The effects of uncorrelated noise decrease with a factor equal to the inverse of the square root of N.

We may extract the phase from the cross-power spectrum

$$\begin{aligned}\theta_{12}(\omega) &= \arg\left(\hat{S}_{12}(\omega)\right) \\ &= k(\omega)(x_2 - x_1)\end{aligned} \tag{4.6}$$

Thus, we may calculate the phase velocity dispersion curve by rearranging Equation 4.6 as follows

$$V_R(\omega) = \frac{\omega(x_2 - x_1)}{\theta_{12}(\omega)} \tag{4.7}$$

Figure 4.8a shows an example of a phase spectrum with a pair of receivers separated by 8 m extracted from the reference dataset presented at the beginning of the chapter. This example is typical of phase spectra measured using a transient source. The large uncertainty in the phase spectrum at high frequencies is likely due to the inability of the source to generate sufficient energy and to geometric attenuation of the waves. The plot is presented as a *wrapped phase* with values of $\theta_{12}(\omega)$ ranging from π to $-\pi$ (i.e., modulo 2π). To apply Equation 4.7, the user must unwrap the phase for the frequency range of interest by adding integer multiples of 2π in the following manner[01]

$$\theta_{\text{unwrap}}(\omega) = \theta_{\text{wrap}}(\omega) \pm 2n\pi, \quad n = 0,1,2\ldots \tag{4.8}$$

Choosing the proper values of n can be a source of uncertainty in SASW tests. This step is often conducted using some automated algorithms (e.g., Poggiagliolmi et al. 1982), but many problems can arise due to the influence of noise that can prevent the correct identification of multiple integer cycles and can produce fictitious jumps in the wrapped phase (see, e.g., Figure 4.9). Operator judgment may solve this problem, but it still remains a subjective procedure (Al-Hunaidi 1992).

Other spectral quantities can help in defining the frequency range in which the information is reliable. The autopower spectra at the two receivers (Figure 4.8c and d) provide an indication of energy distribution. Where their values are high, frequency ranges are likely to be those in which the generated signal is stronger; hence, the latter prevails over uncorrelated noise. This is the reason why light sources (e.g., a small hammer) are used for small receiver spacing because these sources generate much more energy at a higher frequency. Conversely, heavy sources (e.g., a massive weight

[01] Negative values of θ are often used to represent a phase due to propagation delay. It may be necessary to multiply Equation 4.8 by –1.

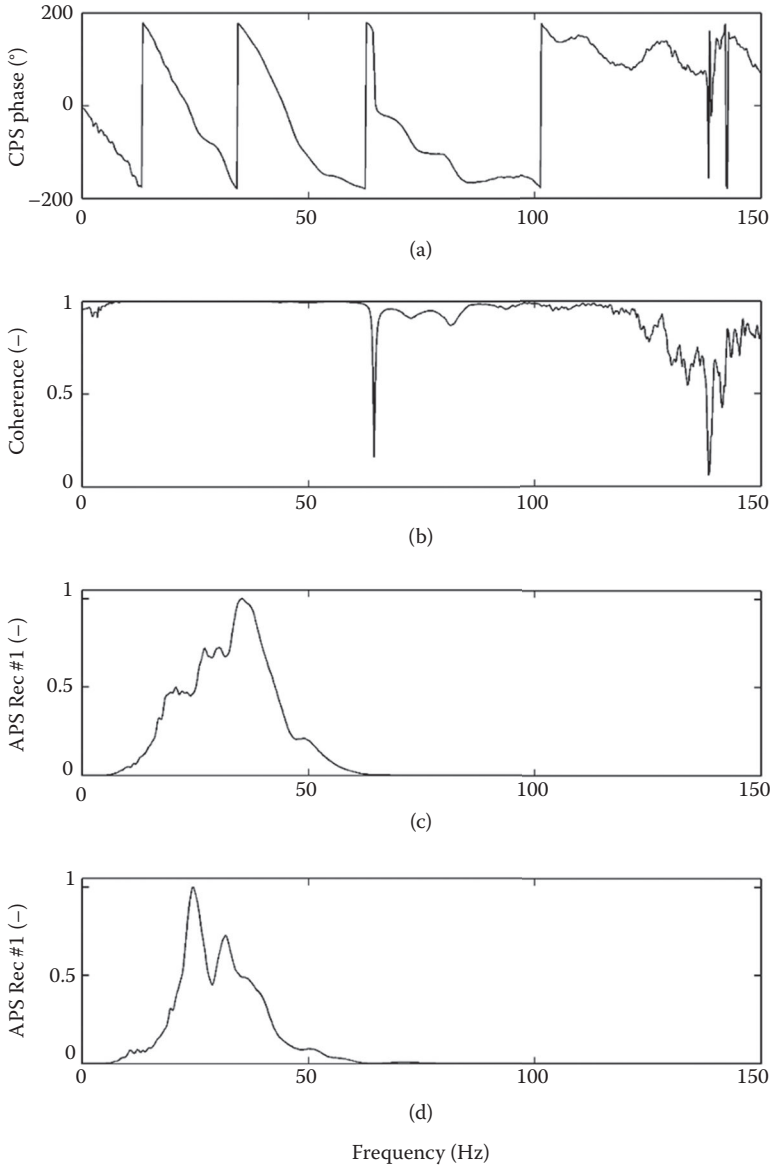

Figure 4.8 Example of spectral quantities for the interpretation of a two-receiver SASW test (source: 130 kg weight drop; inter-receiver distance: 8 m): (a) cross-power spectrum (wrapped); (b) coherence function; (c) normalized autopower spectrum (Receiver 1); (d) normalized autopower spectrum (Receiver 2).

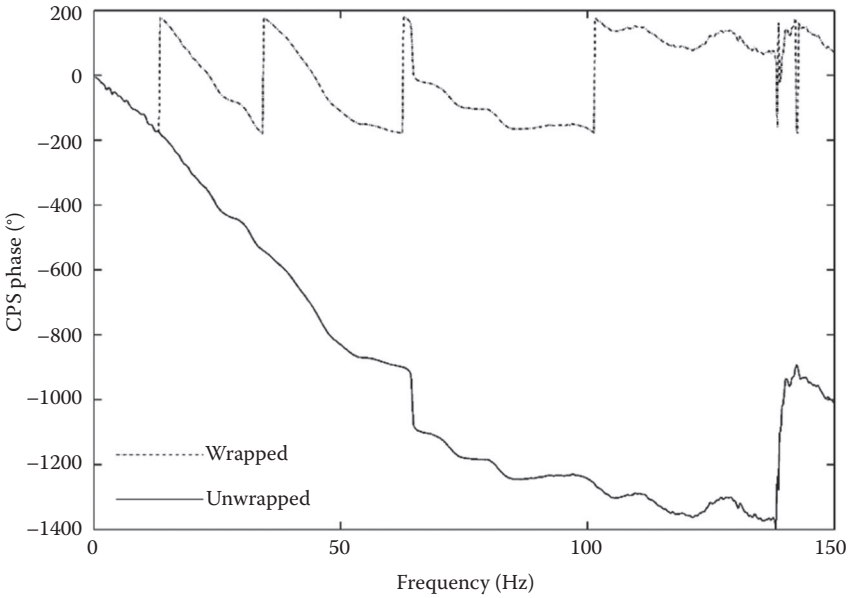

Figure 4.9 Phase unwrapping for the cross-power spectrum of Figure 4.8a. At about 140 Hz, the anomaly represents a typical example of failure of the unwrapping procedure in identifying phase jumps.

drop system) generate strong signals with low-frequency content; hence, they are used for wide spacings. For the example of Figure 4.8, the auto-power spectra indicate that the energy is concentrated in the frequency range between about 10 and 60 Hz. The assessment of signal quality is made using the "coherence function," namely a spectral quantity obtained comparing different registrations that is a measure of the degree by which input and output signals are linearly correlated. A value close to unity is an index of good correlation; hence, the recorded signals can be considered genuine and unaffected by ambient noise. The coherence function between two receivers, γ_{12}, is defined as

$$\gamma_{12}^{2}(\omega) = \frac{\hat{S}_{12}(\omega) \cdot \overline{\hat{S}_{12}(\omega)}}{S_{11}(\omega) \cdot S_{22}(\omega)} \tag{4.9}$$

where the upper bar denotes the complex conjugate. The ordinary coherence function provides a measure of how the measured particle motion at the first receiver is related to the particle motion recorded at the second receiver.

It can be shown that $0 \leq \gamma^2_{r_1 r_2} \leq 1$, with the upper bound $\gamma_{r_1 r_2} = 1$ corresponding to a situation where there is an exact linear relationship between the signals recorded at the two receivers (Bendat and Piersol 2010). Low coherence values may be attributed to the presence of noise or, more generally, to the situation where the measured particle motion at the two receivers is not linearly related. An additional cause for observed low values of the coherence function may be the near-field effect (Lai 1998). In general, using a large number of shots in evaluating the average spectra results in a prevalence of the signal over uncorrelated noise and hence in a high value of the coherence function. For example, the coherence function of Figure 4.8b has been estimated considering six shots. It shows that the signals are linearly correlated up to about 60 Hz, which is the range over which the source provides most of the energy (as confirmed by the autopower spectra at the two receivers reported in Figure 4.8c and d).

On site, the visual inspection of the aforementioned spectral quantities can provide a direct assessment of the data quality, helping in judging if it is necessary to collect additional data.

The dispersion curve corresponding to the phase spectrum of Figure 4.9 is shown in Figure 4.10. Note that only frequencies between 12 and 55 Hz have been included considering the frequency range with high signal-to-noise ratio and the limitations related to near-field effects, which will be discussed hereafter.

Information for a limited frequency range is typically obtained with a given receiver spacing. This is due to many factors such as wave attenuation, spatial aliasing, near-field effects (see the discussion at the end of

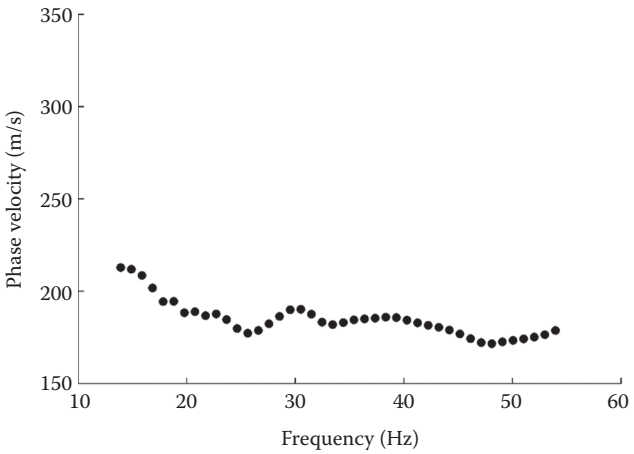

Figure 4.10 Experimental dispersion curve estimated from the phase of the cross-power spectrum reported in Figure 4.9 (receiver spacing: 8 m).

this section). If data at other frequencies are required, the test may be repeated using different receiver spacing according to the measurement schemes of Figure 4.7. The common receivers midpoint scheme is preferable because of its symmetry. Moreover, the source may also be moved to the opposite side of the array to determine the dispersion curve in the "reverse" direction in an effort to average any differences in phase velocity due to lateral inhomogeneity at the site and/or internal phase distortions in receiver responses. The common source array is easier to be implemented in the field, especially if heavy sources are used.

In usual practice, the receiver spacing is doubled in the new setup, implementing a geometrical progression. Dispersion curves from individual receiver spacings are combined to form a composite dispersion curve for the site.

The SASW technique has several important limitations that arise from the use of only two receivers. First, spatial aliasing complicates interpretation of the test results. To avoid spatial aliasing, the wavenumber must be less than the Nyquist wavenumber. In the context of the SASW test

$$\begin{aligned} k(\omega) &\le k_{\text{Nyquist}} \\ \frac{\theta_{12}(\omega)}{x_2 - x_1} &\le \frac{\pi}{x_2 - x_1} \\ \theta_{12}(\omega) &\le \pi \end{aligned} \tag{4.10}$$

Thus, the "jumps" in the phase spectrum and, more specifically, the need to choose n in Equation 4.8 are the result of spatial aliasing. As noted earlier, this aspect of SASW test interpretation often results in uncertainty. Although "automatic" phase unwrapping algorithms are available, they are frequently fooled by poor quality data, and manual interpretation of the phase spectrum is usually required.

Second, the wavenumber resolution is generally poor for receiver spacings normally used in SASW tests. The main lobe in the aperture smoothing function (see Chapter 3) is wide and spans the entire range of wavenumbers defined by $-k_{\text{Nyquist}} \le k \le k_{\text{Nyquist}}$. The most important practical implication of the poor wavenumber resolution is that it is impossible to resolve multiple surface wave modes in the SASW test. The smoothing produced by convolving the aperture smoothing function with multiple modes propagating at different wavenumbers yields only a single apparent wavenumber. Indeed, the procedure based on the phase of the cross-power spectrum (Equation 4.7) yields a single value of the time delay and hence of phase velocity for any testing setup, with no possibility to discriminate different modes of propagation.

Another main issue is related to near-field effects (i.e., deviations from the usual assumption that the propagation is dominated by plane Rayleigh waves). Indeed, the influence of body wave components may affect

the motion detected at the first receiver, and the consequent distortion leads to an underestimation of the actual phase velocity.

Initially, it was proposed to discard all data for which the distance between the source and the first receiver was less than one-third of the obtained wavelength (Heisey et al. 1982). However, some numerical analysis showed that this criterion underestimates the extension of the near-field (Sànchez-Salinero 1987).

Usually, a near-field extension of a half wavelength is assumed for a normally dispersive soil (i.e., where stiffness is increasing with depth), whereas about two wavelengths is a more prudential estimate for a strongly inverse dispersive soil (i.e., where a soft layer is present below or is trapped between stiffer ones). These indications are based on numerical simulation of the complete wave field generated by a point source for different layer configurations (Tokimatsu 1995). Nevertheless, there still is an open discussion about the influence of direct and reflected/refracted body waves on the recorded signals.

It is important to remark that because the Rayleigh waves are dispersive in layered media, the distance at which the condition of far-field can be assumed is not a constant but depends on the frequency analyzed. For example, considering a wrapped cross-power spectrum phase (Figure 4.8a), discarding data affected from a near-field that is a half wavelength long is equivalent to cutting information given by an initial section of 180° (i.e., the portion between 0 and the first jump).

Finally, because of attenuation, data relative to inter-receiver distance higher than about three wavelengths are usually strongly affected by noise (Stokoe et al. 1988); therefore, it is preferable to discard them.

In summary, assuming that no strong contrasts of stiffness are present, for each receivers configuration, the following restrictions are applied to select reliable dispersion curve data points

$$\frac{D}{3} < \lambda < 2D \tag{4.11}$$

in which λ is the estimated wavelength and D is the inter-geophone distance that is taken equal to the distance source-first geophone. A visual interpretation of these filtering criteria is given in Figure 4.11, showing the whole set of data that can be obtained by a given experimental setup and the subset that is considered acceptable.

As seen here, the construction of the experimental dispersion curve is strongly affected by the operator's experience because a selection of significant and corrected data is actually required. Moreover, because this is time-consuming and involves many manipulations of the data, it is usually conducted in the office after data collection in the field.

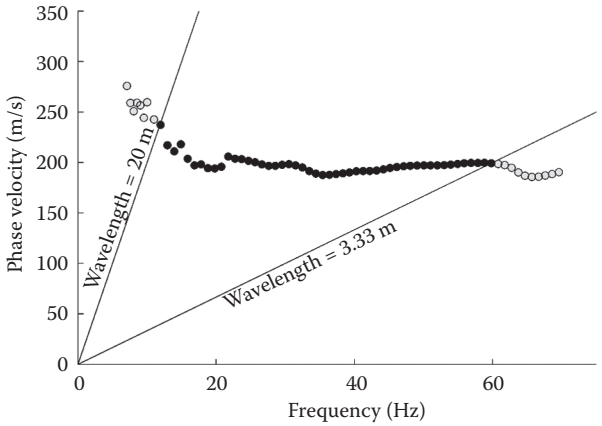

Figure 4.11 Example of application of filtering criteria to the dispersion curve corresponding to one source–receiver configuration with 10 m spacing; only the data points between the two limiting wavelengths are considered acceptable (full black dots).

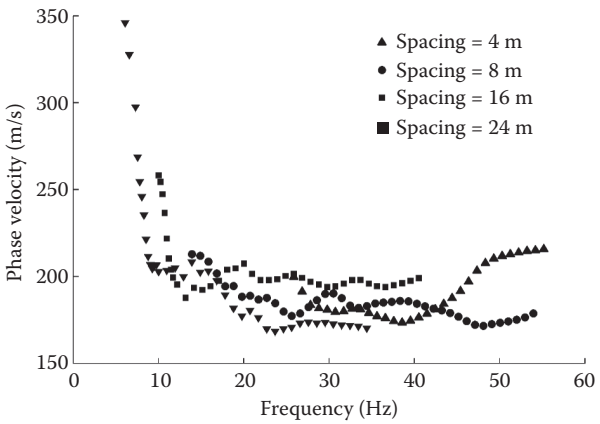

Figure 4.12 Example of experimental dispersion curve estimated with four testing setup in SASW test.

The whole ensemble of data collected using a series of geophone configurations (Figure 4.12) has to be assembled to create a single global dispersion curve covering a wide enough range of frequency. This is necessary because the automated inversion algorithms (see Chapter 6) have to compare the numerical dispersion curve to the experimental one by computing a norm of the distance between the two. Usually for any subset in frequency

(or wavelength), the average value of phase velocity is assigned to the central frequency (or wavelength) of the subset. Still, there are a few problems—one is that the choice of reducing points in the frequency or wavelength domain must be made consistently with the domain successively adopted for the inversion process. The other one is about the significance of adopting the mean value of a population that hardly can be seen as a statistical distribution (this because the number of overlapping information in a given frequency range is arbitrary, depending on testing configurations, quality of data, etc.).

4.4 MULTI-OFFSET PHASE ANALYSIS

MOPA is a surface wave analysis technique in which the phase versus offset of particle motion is processed to estimate the phase velocity (Strobbia and Foti 2006). It can be considered an extension of the SASW method for multichannel arrays. The procedure also allows for the identification of lateral variations and a robust estimation of the velocity. Moreover, traces can be weighted as a function of their uncertainty, thus improving the accuracy.

The algorithm is based on the fact that the surface wave displacement in frequency domain, in a laterally homogeneous medium and in far offset from the source, can be written as modal summation, separating the frequency-dependent and offset-dependent terms (see also Section 2.4.2)

$$s(\omega,x)=\sum_m I(\omega)R_m(\omega)\frac{e^{-\alpha_m(\omega)x}}{\sqrt{x}}e^{i(\omega t-k_m(\omega)x+\ _0(\omega))} \tag{4.12}$$

where $I(\omega)$, $R_m(\omega)$, $\alpha_m(\omega)$, and $\varphi_0(\omega)$ are the amplitude spectrum of the source, the site response for mode m, the intrinsic attenuation for mode m, and the phase spectrum of the source, respectively.

When a single mode is present or dominant, the amplitude and the phase can be represented as

$$A(\omega,x)=I(\omega)R(\omega)\frac{e^{-\alpha(\omega)x}}{\sqrt{x}} \tag{4.13}$$

and

$$\varphi(\omega,x) = -k(\omega)\cdot x+\varphi_0(\omega) \tag{4.14}$$

Hence, for a single mode and a single frequency, the amplitude and the phase are an exponential and a linear function of offset from the source, respectively (Figure 4.13).

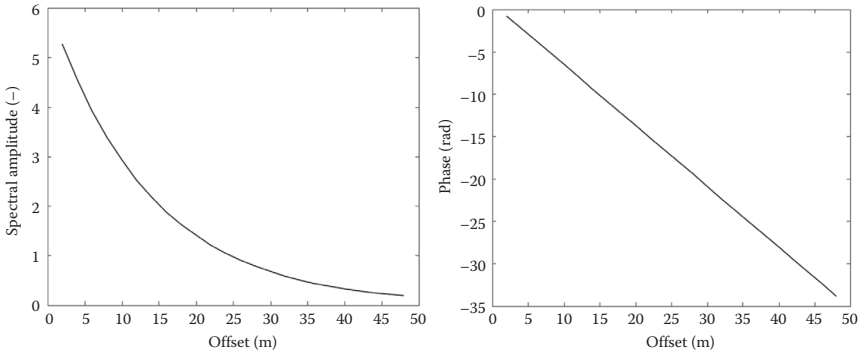

Figure 4.13 Theoretical behavior of the phase and the amplitude for one frequency as a function of the offset.

The extraction and inspection of the phase in experimental data is the key step of MOPA. It allows the estimation of the phase velocity as well as the identification of lateral variations and of near-field effects.

Figure 4.14 shows the reference shot gather together with the corresponding seismograms after narrow-band filtering at 9 and 19 Hz, respectively. The filtered seismograms show graphically the velocity of the propagating energy at the considered frequency. This experimental evidence can be implemented in an efficient processing procedure.

The time shift of a harmonic signal is related to its phase

$$s(t) = \sin(\omega t + \varphi) = \sin(\omega(t - \Delta t)) \tag{4.15}$$

gives

$$\varphi = -\omega \cdot \Delta t = -2\pi f \cdot \Delta t \tag{4.16}$$

In practice, it is possible to estimate the time shift of the harmonic (stationary and infinite) function from the phase at a given frequency. The phase can be extracted using the Fourier transform. Figure 4.15a and b report the amplitude and the phase, respectively, for the 19 Hz frequency of the reference dataset (Figure 4.15c).

In Figure 4.15b, the phase is reported in a modulo-2π representation and needs to be unwrapped to describe the linear propagation along the array. The simplest unwrapping procedure changes absolute jumps greater than π radians to their complements to 2π. Figure 4.16 shows the wrapped phase and its corresponding unwrapped phase.

The unwrapped phase is directly related to the time delay for the considered frequency. In the example of Figure 4.17, considering the 46 m

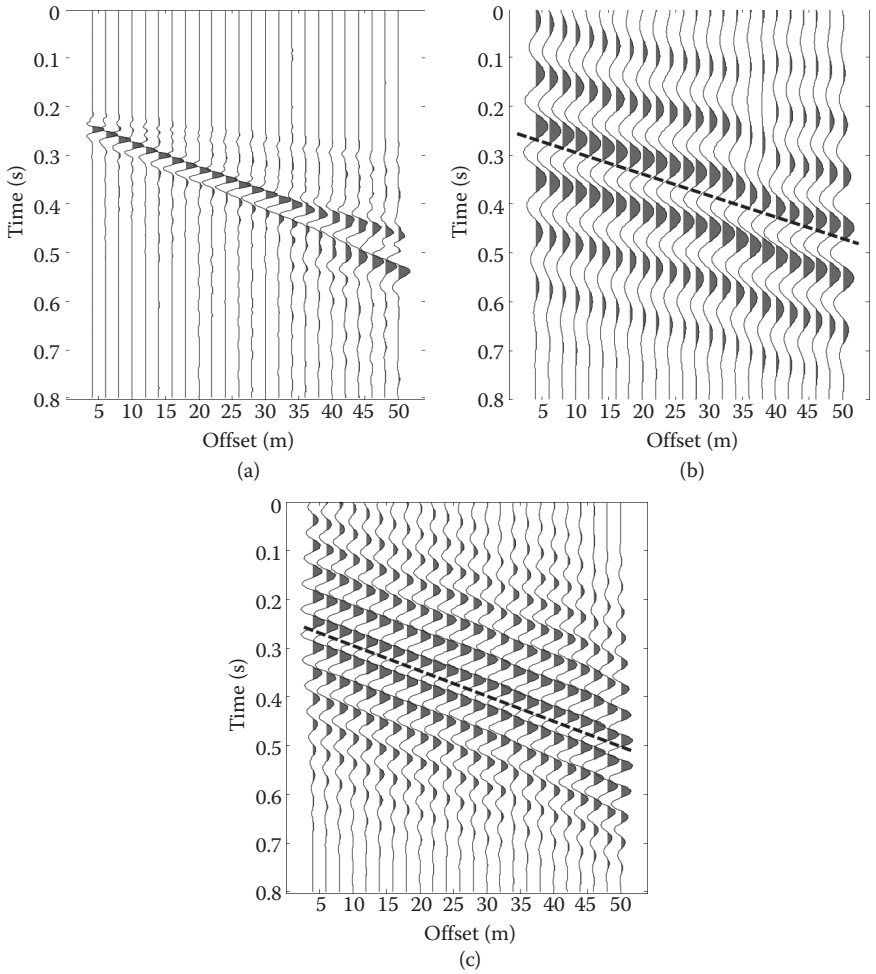

Figure 4.14 (a) Raw seismogram (b) band-pass filtered around 9 Hz, and (c) around 19 Hz. The slope in time-offset represents the phase velocity at the considered frequency.

interval between the first receiver and the last receiver, the phase difference of about 28 radians gives a time delay of 0.236 s and a phase velocity of 195 m/s.

If multiple shots with the same receiver spread are acquired, the effect of the incoherent noise on the data can be assessed with a rigorous statistical approach. For example, Figure 4.18a shows the traces recorded at the 10th geophone for all the available shots in the reference dataset. In Figure 4.18b, the individual phase values at the frequency of 19 Hz are

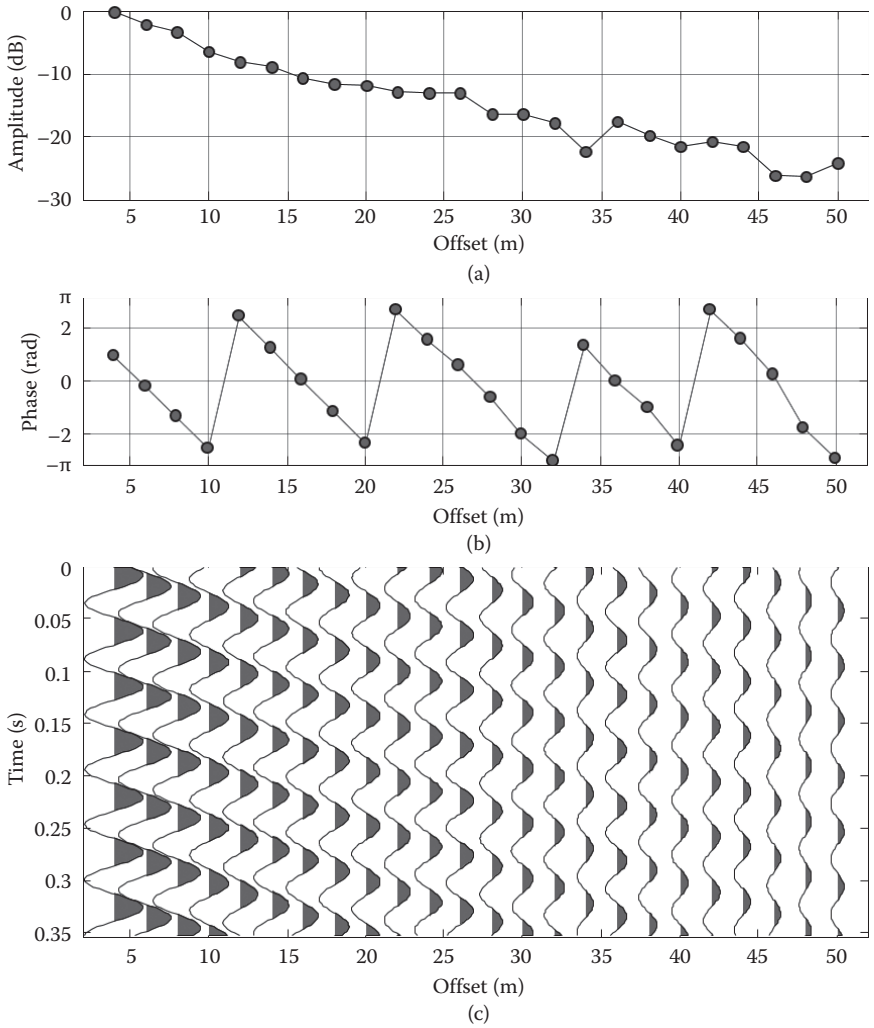

Figure 4.15 (a) Amplitude and (b) phase of the 19 Hz harmonic as a function of offset for the reference dataset. The corresponding harmonic traces are represented in (c).

represented with the dots and the mean is indicated with the dashed line. The shaded area indicates the interval between the mean and one standard deviation. The error bar in Figure 4.18c summarizes the distribution of the phase at 19 Hz at the considered receiver No. 10.

With a multichannel array, a phase distribution can be obtained for each receiver station, for each considered frequency. Considering three

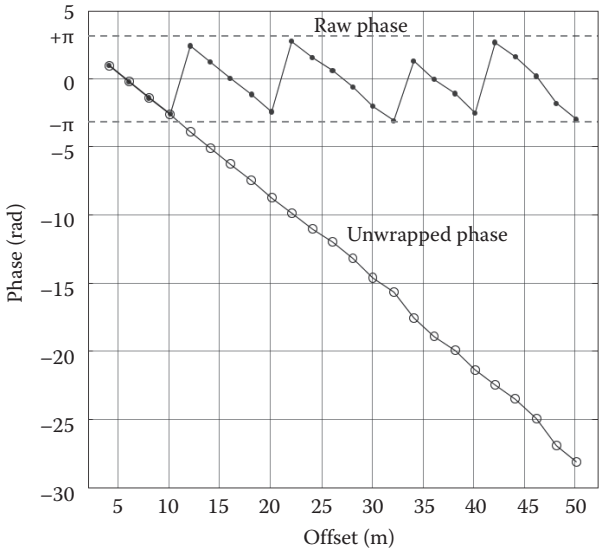

Figure 4.16 Experimental raw phase and unwrapped phase at the frequency of 19 Hz of the spectrum of Figure 4.15b.

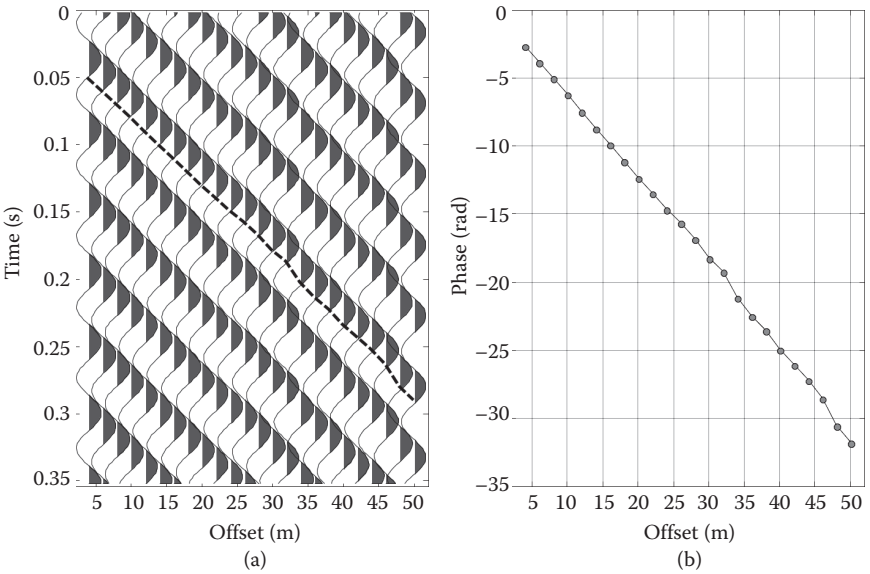

Figure 4.17 (a) Normalized harmonic components at 19 Hz with the annotated velocity and (b) phase versus offset graph for the considered frequency.

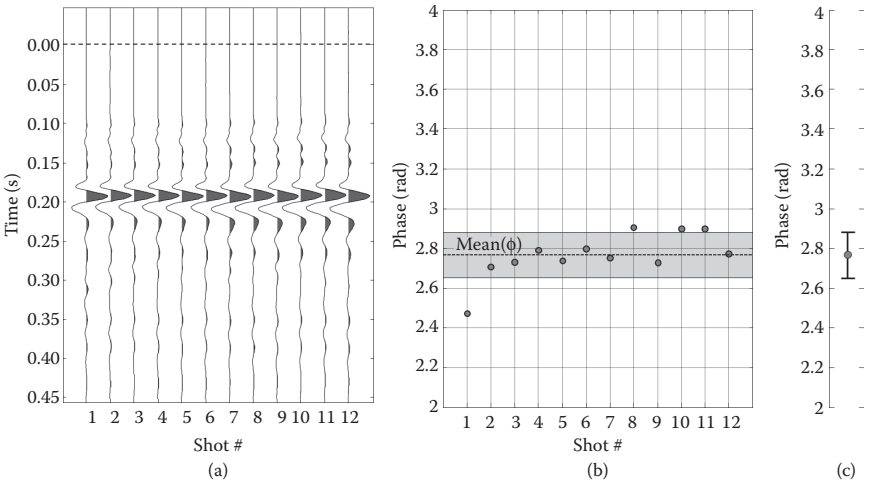

Figure 4.18 Statistical analysis of the phase: (a) traces recorded by the 10th receiver of the array; (b) extracted phase for each shot and the average value (the shaded area represents the $\pm\sigma$ interval); (c) average phase and the corresponding standard deviation plotted as error bar.

frequency values, the phase distribution is plotted in Figure 4.19. Statistical data can be used to obtain the best estimate of the wavenumber at each frequency and its associated standard deviation by using the following procedure.

The linear model describing the phase in a laterally homogeneous medium in the far-field of the source is described by Equation 4.14, which can be written for each receiver as

$$\varphi_i = k \cdot x_i + \varphi_0 \tag{4.17}$$

where the phase φ_i at the offset x_i depends on the unknown wavenumber k. Considering an array with N receivers, we get

$$\begin{cases} {}_1 = k \cdot x_1 + {}_0 \\ {}_2 = k \cdot x_2 + {}_0 \\ \quad\vdots \\ {}_N = k \cdot x_N + {}_0 \end{cases} \tag{4.18}$$

that in matrix form can be expressed as

$$\boldsymbol{\Phi} = \boldsymbol{G} \cdot \boldsymbol{M} \tag{4.19}$$

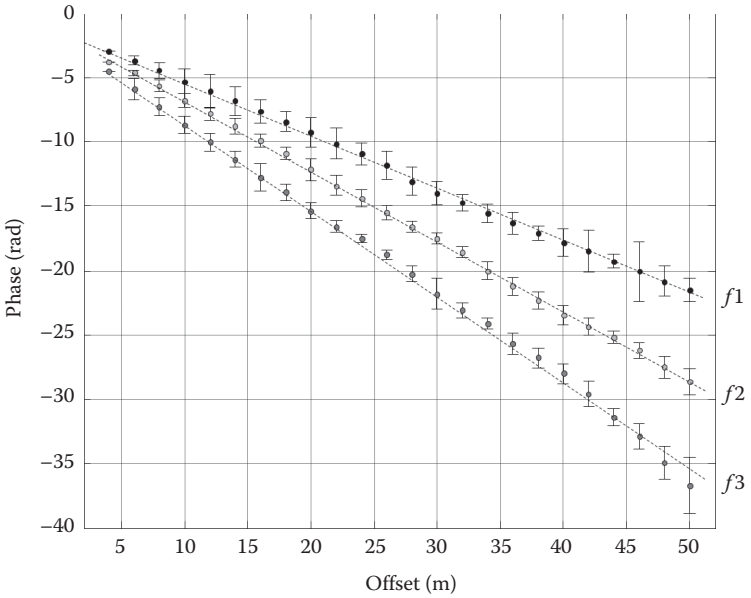

Figure 4.19 Phase distribution versus offset for three different frequencies accounting for the standard deviation of the experimental data.

where $\mathbf{\Phi} = [\varphi_1, \varphi_2, \ldots, \varphi_N]^T$ is the vector of the experimental phases, $\boldsymbol{M} = [k, \varphi_0]^T$ is the vector of the unknown model parameters, and $\boldsymbol{G} = \begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_N & 1 \end{bmatrix}$ is the data kernel matrix, depending on the geometry (Menke 1989).

The model parameters can be estimated in the least-squares sense by the pseudoinverse $\boldsymbol{G}^{-g}$

$$\boldsymbol{M} = \boldsymbol{G}^{-g} \cdot \mathbf{\Phi} \tag{4.20}$$

with $\boldsymbol{G}^{-g} = (\boldsymbol{G}^T \cdot \boldsymbol{G})^{-1} \cdot \boldsymbol{G}^T$.

The experimental values of the phase can be weighted according to the corresponding standard deviation in order to estimate the uncertainty of the estimated wavenumber. The weighted least-squares estimation can be obtained by replacing the pseudoinverse with the corresponding weighted version $\boldsymbol{G}_W^{-g}$, written as (Tarantola 2005)

$$\boldsymbol{G}_W^{-g} = \left(\boldsymbol{G}^T \cdot \boldsymbol{W}^T \cdot \boldsymbol{W} \cdot \boldsymbol{G}\right)^{-1} \cdot \boldsymbol{G}^T \cdot \boldsymbol{W}^T \cdot \boldsymbol{W} \tag{4.21}$$

where W is the diagonal matrix containing the weights (i.e., the standard deviation of the phase).

With the assumption of normally distributed and independent data, the linear relationship between the experimental phases and the model parameters

$$\boldsymbol{M} = \boldsymbol{G}_{\overline{W}}^{-g} \cdot \boldsymbol{\Phi} \tag{4.22}$$

implies that

$$\boldsymbol{\sigma}_M^2 = \boldsymbol{G}2_{\overline{W}}^{-g} \cdot \boldsymbol{\sigma}_\Phi^2 \tag{4.23}$$

where $\boldsymbol{\sigma}_M^2$ and $\boldsymbol{\sigma}_\Phi^2$ are the covariance matrices of the model parameters and of the data, respectively, and $G2_{\overline{W}}^{-g}$ is the matrix containing the squares of the elements of $G_{\overline{W}}^{-g}$ (Santamarina and Fratta 2010).

Adopting this procedure, the mean value of the wavenumber and the associated standard deviation can be estimated at each frequency of interest. The wavenumber is then used to compute the phase velocity V, the standard deviation of which can be estimated with a first-order approximation

$$V = \frac{2\pi \cdot f}{k} \tag{4.24}$$

$$\sigma_V = \frac{2\pi \cdot f}{k^2} \sigma_K \tag{4.25}$$

Figure 4.20 shows the mean and standard deviation of the unwrapped phase for the reference dataset.

Figure 4.21 shows the experimental dispersion curve with the associated standard deviation. However, the standard deviation does not describe the shape of the probability density function of the phase velocity.

This procedure also allows an assessment of the linearity of the phase in the experimental dataset. Indeed, as discussed earlier, the phase is a linear function of the offset only in a laterally homogeneous medium and in the far-field of the source. These hypotheses are not always verified. Indeed, real data may be recorded at sites where lateral variations are present, and for a given array and source configuration, some of the receivers may fall in the near-field, for some frequencies.

For example, near-field effects can be detected in the reference dataset. Figure 4.22 shows amplitude and phase as a function of offset, for the frequency of 5.9 Hz. The corresponding wavelength is about 33 m, and despite the low standard deviations, the phase of the first six traces appears anomalous with respect to the trend.

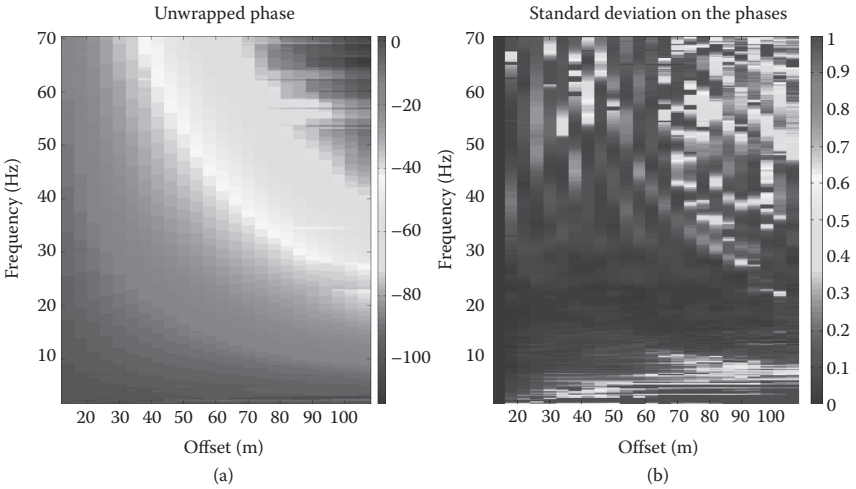

Figure 4.20 (a) Mean and (b) standard deviation of the phase as a function of offset and frequency, for the reference dataset.

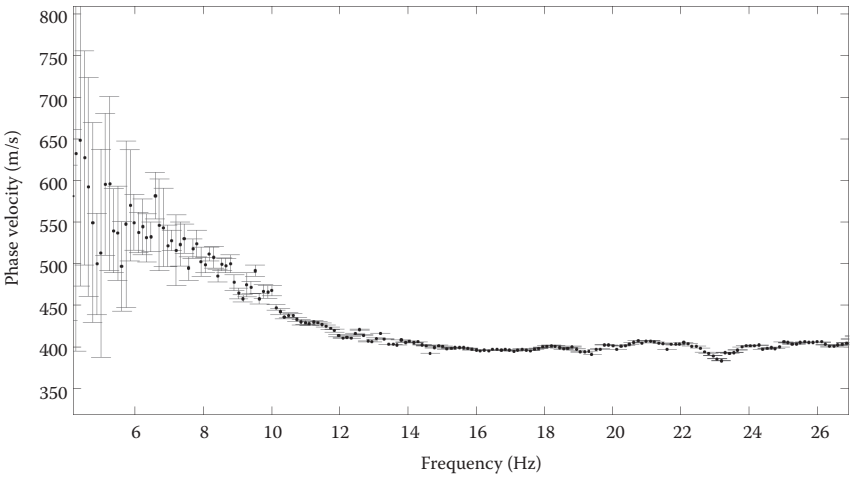

Figure 4.21 Dispersion curve obtained with MOPA. The dispersion curve is obtained from the best estimate, and the experimental uncertainties are statistically estimated.

It is clear that, at an offset of about 16 m, the slope of the phase versus offset changes. It can also be observed that, approximately at the same offset, the attenuation also has a large variation. This anomalous behavior is likely to be related to the near-field effect. Indeed, observing the behavior at different frequencies, it can be found that the distance at which

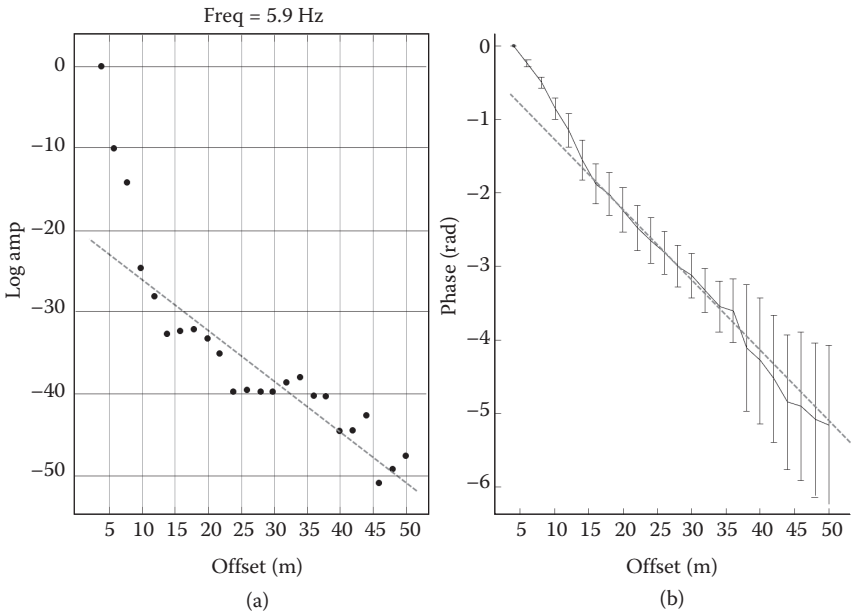

Figure 4.22 (a) Amplitude after geometric spreading correction and (b) unwrapped phase versus offset at the frequency of 5.9 Hz, for the reference dataset.

the linearity is lost changes with the wavelength. When lateral variations are present at the site, the change of slope tends to be located at a specific position along the array.

The identified near-field region can be removed from the data; at every frequency, only the traces below a near-field criterion, i.e. the traces with an acceptable linearity, are considered.

Similarly, the deviation from the theoretical linear behavior can indicate the presence of lateral variations. An example of severe lateral variation is provided in Figure 4.23. The shot gather is plotted in Figure 4.23a and the phase versus offset in Figure 4.23b.

A simple visual inspection of the data and of the phase plots can help with identifying the lateral variations. However, a more rigorous procedure can be implemented; given the estimated phase uncertainties, the linearity of the phase can be statistically tested. The principle is that if the uncertainty on the different phase points is low, the regression misfit cannot be too large; a large misfit implies that the model is not linear.

If each of the N individual prediction errors is assumed to be a Gaussian independent random variable x, each with zero mean and unit standard

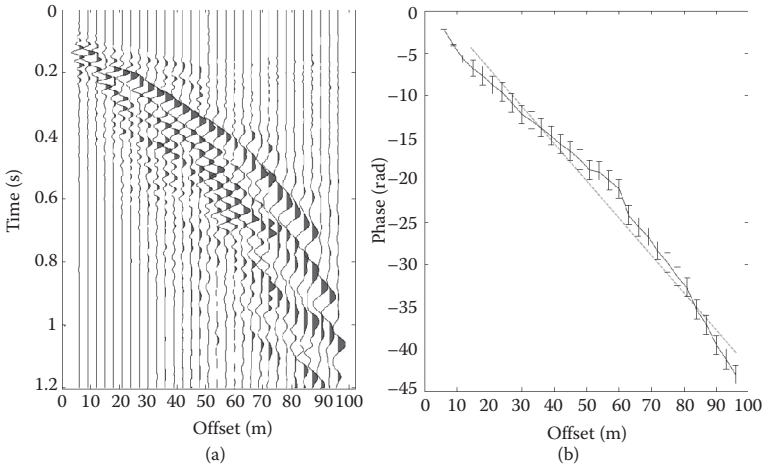

Figure 4.23 Example of data with lateral variations: (a) shot gather; (b) phase distribution for a constant frequency.

deviation, the sum of their squares is a random variable with chi-squared probability density with N degrees of freedom

$$\chi^2 = \sum_{i=1}^{N} x_i^2 \tag{4.26}$$

Therefore, the total misfit in the least-squares sense has a chi-squared distribution. If the different errors have different standard deviations, the overall misfit can be predicted by weighting with the covariance matrix C and including the vector of the means μ, to give a random variable with the same probability density

$$\chi^2 = (x - \mu)^T C(x - \mu) \tag{4.27}$$

Given the best fitting linear trend, the corresponding total misfit is compared with the misfit distribution predicted from the phase uncertainties. A 95% confidence criterion can be adopted in the chi-squared test of the final misfit, to identify automatically the presence of nonlinearity.

If lateral variations are identified in the data, a single dispersion curve cannot be reliably extracted. A possible approach for identifying automatically sharp lateral variations and for handling cases such as the one in Figure 4.23 has been proposed by Vignoli and Cassiani (2009). The phase distribution is analyzed to identify the consistent phase knee-points, and the data are split into subsets where the linearity hypotheses are verified. A continuous analysis of the local phase gradient to extract the local wavenumber with an extension of the MOPA procedure has been proposed by Vignoli et al. (2011).

This technique may also be easily adapted to simultaneously determine the dispersion and attenuation curves by using the magnitude and phase of the cross-power spectrum to determine a complex-valued wavenumber at each frequency (see Section 5.3).

4.5 SPATIAL AUTOCORRELATION

The SPAC method was developed by Aki (1957, 1965) to calculate dispersion curves using microtremor data. The original implementation of the method used a circular receiver array with an additional receiver located at the center of the circle. We will take the more general approach of the extended spatial autocorrelation (ESAC) method (Ohori et al. 2002), which does not impose such restrictions on the geometry of the array.

4.5.1 Single source

The basis of the method is the observation that the spatial autocorrelation function may be obtained via an inverse spatial Fourier transform of the wavenumber

$$\begin{aligned} R(\chi) &= E\left[s(\mathbf{x},t)s(\mathbf{x}+\chi,t)\right] \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\mathbf{k})\, e^{-i\mathbf{k}\cdot\chi} d\mathbf{k} \end{aligned} \tag{4.28}$$

where χ is the spatial lag and $E[..]$ denotes the expected value of a quantity. Considering a unit-amplitude plane wave described by

$$s(\mathbf{x},t) = e^{i(\omega_0 t - \mathbf{k}_0\cdot\mathbf{x})} \tag{4.29}$$

the wavenumber spectrum of the signal is

$$S(\mathbf{k}) = \delta^2(\mathbf{k} - \mathbf{k}_0) \tag{4.30}$$

For the sake of compactness, frequency dependency is omitted reporting the expressions for a given frequency ω_0.

Substituting Equation 4.30 into Equation 4.28 yields

$$\begin{aligned} R(\chi) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \delta^2\left(\mathbf{k}-\mathbf{k}_0\right) e^{-i\mathbf{k}\cdot\chi} d\mathbf{k} \\ &= \frac{1}{2\pi} e^{-i\mathbf{k}_0\cdot\chi} \\ &= \frac{1}{2\pi}\left[\cos\left(\mathbf{k}_0\cdot\chi\right) - i\sin\left(\mathbf{k}_0\cdot\chi\right)\right] \end{aligned} \tag{4.31}$$

Because the autocorrelation function is an even function of the spatial lag, we can limit our interest to the real part of the solution. Thus

$$R(\chi) = \frac{1}{2\pi}\cos\left(\mathbf{k}_0 \cdot \chi\right) \tag{4.32}$$

Finally, the autocorrelation function may be normalized to obtain the autocorrelation coefficient:

$$\begin{aligned}\rho(\chi) &= \frac{R(\chi)}{R(\chi = 0)} \\ &= \cos\left(\mathbf{k}_0 \cdot \chi\right)\end{aligned} \tag{4.33}$$

Equation 4.35 may be used to estimate the wavenumber (and thus phase velocity) from observations of the spatial autocorrelation coefficient of the wave field. The most efficient means of calculating experimental spatial autocorrelation coefficients is via the normalized cross-power spectra between signals at two receiver positions

$$\rho(\chi,\omega) = \frac{\Re\left(\hat{S}_{ij}(\omega)\right)}{\sqrt{\hat{S}_{ii}(\omega)\hat{S}_{jj}(\omega)}} \tag{4.34}$$

where $\hat{S}_{ij}(\omega)$ is the ensemble average cross-power spectrum between receivers i and j, and $\hat{S}_{ii}(\omega)$ and $\hat{S}_{jj}(\omega)$ are the autopower spectra at the two receivers.

Figure 4.24 shows examples of the use of the spatial autocorrelation method to determine the dispersion curve from an active surface wave test using a linear array of receivers. In this case, the wavenumber and spatial lag may be represented by scalar quantities. The resulting dispersion curve is shown in Figure 4.25.

A major concern in SPAC analysis of active-source data is that the signal for the evaluation of the experimental autocorrelation coefficients at the reference receiver may be affected by near-field effect, which is not accounted for in the formulation. Moreover, the method provides the estimate of a single phase velocity; hence, it does not allow for the separation of modes of propagation, providing an apparent phase velocity irrespective of the wavenumber resolution.

4.5.2 Isotropic wave field

It may be of interest to consider an *isotropic wave field*, which is comprised of random waves propagating in all possible directions with equal probability. This ideal situation may be reasonably assumed to represent a passive

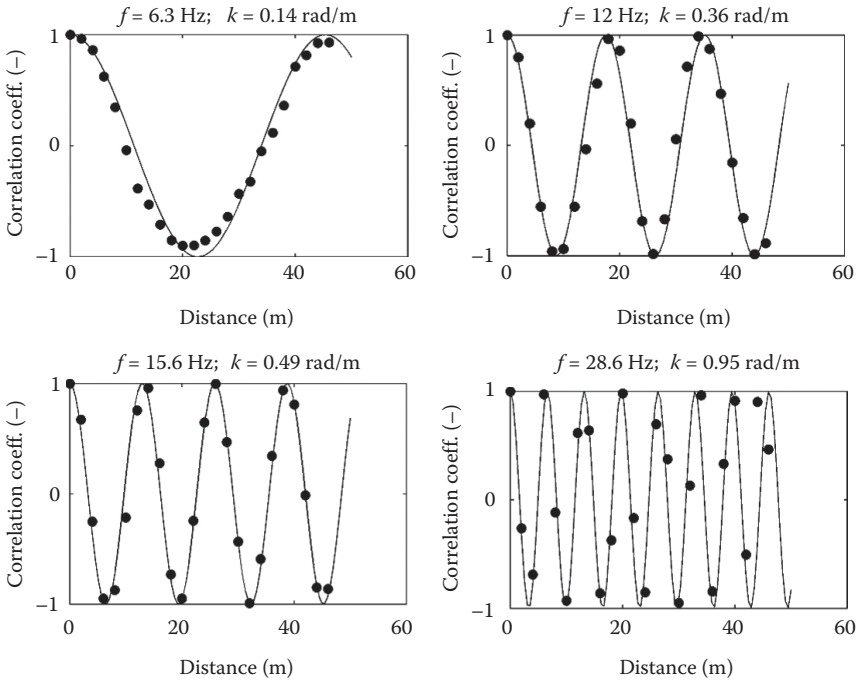

Figure 4.24 Example of fitting obtained at different frequencies with SPAC analysis on active-source multistation data for the reference dataset.

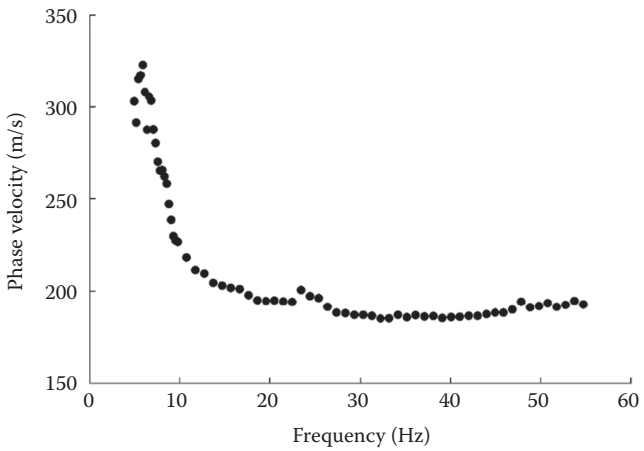

Figure 4.25 Example of experimental dispersion curve estimated with SPAC procedure on active-source multistation data.

surface wave test with multiple, spatially distributed sources. It requires measurements with a two-dimensional (2D) array of sensors on the ground surface. This problem naturally lends itself to polar coordinates (r,ϕ) where

$$r = \frac{x}{\cos(\phi)} = \frac{y}{\sin(\phi)}$$
$$k = \frac{k_x}{\cos(\phi)} = \frac{k_y}{\sin(\phi)} \tag{4.35}$$

and the signal may be represented by $s(r, t)$. The spatial autocorrelation function for this radially symmetric problem becomes (Bracewell 1978)

$$R(r) = E\left[s(0,t)s(r,t)\right]$$
$$= \int_0^\infty S(k) J_0(kr) k \, dk \tag{4.36}$$

which is an inverse Hankel transform where J_0 is the zero-order Bessel function of the first kind.

Considering a unit-amplitude, isotropic wave field is described by

$$s(r,t) = e^{i(\omega_0 t - k_0 r)}, \tag{4.37}$$

its wavenumber spectrum is given by

$$S(k) = \delta(k - k_0) \tag{4.38}$$

(Again, note that for the sake of compactness of the equation, the frequency dependency is omitted and the spectrum is referred to a given frequency ω_0).

Substituting Equation 4.38 into Equation 4.36 yields

$$R(r) = \int_0^\infty \delta(k - k_o) J_0(kr) k \, dk$$
$$= J_0(k_0 r) k_0 \tag{4.39}$$

As noted previously, the autocorrelation function may be normalized to obtain the autocorrelation coefficient

$$\rho(r) = \frac{R(r)}{R(r=0)}$$
$$= J_0(k_0 r) \tag{4.40}$$

To comply with the assumption of radial symmetry, it is necessary to calculate the experimental autocorrelation coefficient using one receiver as a reference (i.e., the receiver at the "origin" of the array). Thus

$$\rho(r,\omega)=\frac{\Re\left(\hat{S}_{0j}(\omega)\right)}{\sqrt{\hat{S}_{00}(\omega)\hat{S}_{jj}(\omega)}} \tag{4.41}$$

where the subscript 0 denotes the reference receiver.

In his original implementation of the method, Aki (1957, 1965) obtained Equation 4.40 by taking the azimuthal average of the spatial autocorrelation function, although this required the use of a circular array (i.e., constant r) as noted previously. The current implementation has no such requirement. Equations 4.33 and 4.40 may be used to determine the wavenumber by minimizing the least-squares error between the theoretical and experimental autocorrelation coefficients as a function of x and r, respectively. A limitation of the SPAC method is that it may be used only for unidirectional (Equation 4.33) or isotropic (Equation 4.40) wave fields (Horike 1985). For active tests, this poses no problem because the user creates unidirectional waves via a controlled source. For passive tests, however, the wave field consists, in general, of multiple sources and may not conform to either of these conditions.

4.6 TRANSFORM-BASED METHODS

The wave field transformations are basic tools in the seismic data processing (Yilmaz 1987) because they allow the separation and identification of different seismic events. The use of global transforms of the dataset for the analysis of the dispersion has been discussed by Nolet and Panza (1976). By means of the wave field transformations, the experimental data are transformed into images of the dispersion curve of each mode. The experimental data are transformed from the original space–time domain into a different domain (ω–p or f–k) in which the propagation parameters are easily identified as spectral maxima. The formal equivalence of different transformations can be proven considering the mathematical properties of the different transforms (Santamarina and Fratta 2010), and there is practically no difference in the obtained dispersion curves (Foti 2000).

In theory, transform-based methods allow the identification of several distinct Rayleigh modes, which could be highly beneficial for the inversion process. In the transformed domain, modes can often be separated even when their presence is not visually detectable in the untransformed data. Nevertheless, the poor spatial resolution (see Chapter 3) often prevents such a possibility, especially in near-surface applications. Indeed, depending on

the aperture of the array, it may be not possible to clearly identify different modes, and the result of the dispersion analysis is an apparent dispersion curve influenced by several modes of propagation (see Section 3.3).

4.6.1 Frequency–wavenumber domain

The original wave field, detected in the space–time domain, can be decomposed into its components at constant frequency and wavenumber. In this way, the seismic-gather is converted into an image of the energy density as a function of the frequency and of the wavenumber. This transform is often used in seismic processing because it allows for events having different frequencies, wavenumbers, and apparent velocities to be separated and filtered out. Different wave phenomena are separated and can be muted and suppressed (*f–k* dip filtering). In particular, in seismic processing, the ground-roll (i.e., surface wave components in the wave field) is viewed as high-energy coherent noise, and its removal is an important step.

The use of *f–k* transform for the processing of surface waves has been proposed by Nolet and Panza (1976). Once the modal wavenumbers have been estimated for each frequency, they can be used to evaluate the dispersion curve recalling that phase velocity is given by the ratio between frequency and wavenumber. An example of application is shown in the following: a synthetic seismogram in time-offset domain (Figure 4.26a) with a single non-dispersive event and a velocity of 200 m/s is transformed into *f–k* domain; the resulting image (Figure 4.26b) represents the energy density as a function of the frequency and of the wavenumber. A single velocity corresponds to a straight line from the origin; the line slope increases with the velocity.

Using a large number of signals (256), Gabriels et al. (1987) were able to identify six experimental Rayleigh modes for a site, which were then used for the inversion process. The possibility of using modal dispersion curves is a great advantage with respect to methods giving only a single dispersion curve (e.g., the two-station method and multistation data processed with MOPA or SPAC methods) because having more information means a better constrained inversion. Nevertheless, it has to be considered that in standard practice the number of receivers for engineering applications is typically small, and reduced spatial sampling strongly affects the resolution of surface wave tests. Receiver spacing influences aliasing in the wavenumber domain, so that if high-frequency components are to be sought, spacing must be small. On the contrary, the total length of the receiver array influences the resolution in the wavenumber domain (see Chapter 3).

The straightforward path to obtain an *f–k* spectrum is the application of the Fourier transform in both time and space on field data (see Section 3.3.7). Typically, the number of samples in time domain and the length of the recording are sufficiently high to obtain a good resolution in the frequency domain, whereas the number of samples in the space domain is limited,

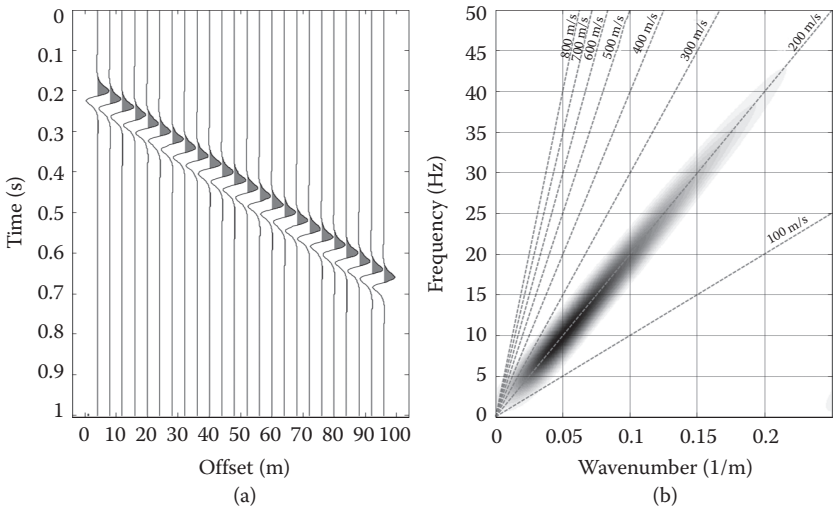

Figure 4.26 Example of synthetic data with a single nondispersive event: (a) synthetic shot gather; (b) frequency–wavenumber spectrum showing a constant phase velocity with frequency.

so that zero padding is applied in space to achieve a reasonable resolution in the wavenumber domain (Figure 4.27a). The space windowing of data is also required to avoid side lobes, which generate local maxima in the spectrum that could be erroneously picked as multiple modes. An example of spectrum computed without tapering is shown in Figure 4.27b. The direct picking of a spectrum with poor spectral resolution, computed without zero padding, is shown in Figure 4.27c.

An example of dispersion analysis in the frequency–wavenumber domain is reported in Figure 4.28, considering the reference dataset. The computation of the spectrum is performed with Hanning tapering and zero-padding in space to a total of 2048 traces, then the spectrum has been computed with a conventional fast Fourier transform algorithm.

The main limitation of using the 2D Fourier transform to directly obtain the f–k spectrum is that traces have to be uniformly spaced. The f–k spectrum for nonequispaced arrays can be evaluated using other techniques of spectral estimation.

The frequency–wavenumber spectrum may be estimated via a process called *frequency-domain beamforming* (Johnson and Dudgeon 1993). The method is conceptually similar to the one based on the τ–p transform (Section 4.6.2).

Let $s(x_m,t)$ be the signal observed at the mth receiver of an array of M receivers. The column vector

$$S(\omega) = [S(x_1,\omega), \ldots, S(x_M,\omega)]^T \tag{4.42}$$

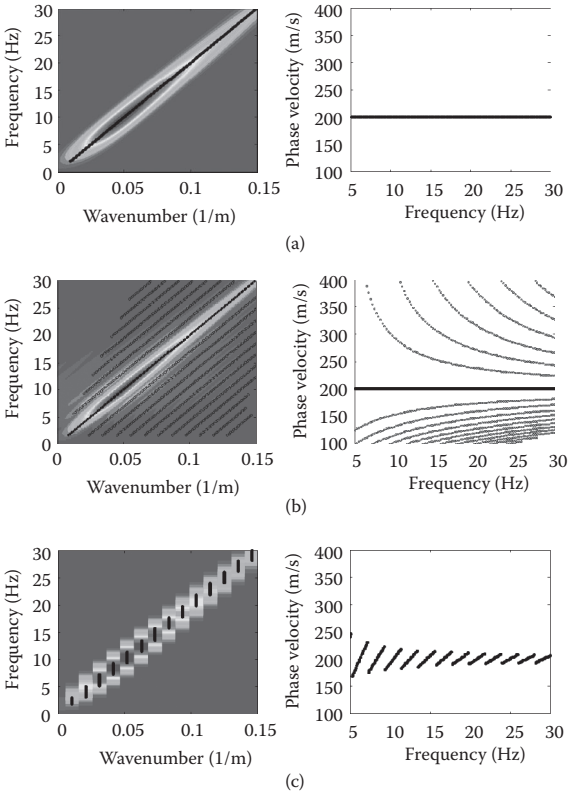

Figure 4.27 Example of *f–k* spectra of the synthetic seismogram (shown in Figure 4.26) and picked dispersion curves. (a) The spectrum (left) and the curve (right) computed with Hanning windowing and zero padding. (b) The spectrum (left) computed without tapering and the curves (right) corresponding to local maxima. (c) The spectrum (left) computed without zero padding and the picked curve (right).

contains the temporal Fourier transform of the signal for each receiver, and

$$\boldsymbol{e}(k) = \left[e^{-ik\cdot x_1}, \ldots, e^{-ik\cdot x_M} \right]^T \tag{4.43}$$

is a *steering vector* (T denotes the transpose of the vector). The output of the conventional frequency-domain beamformer is given by (Johnson and Dudgeon 1993)

$$Z(\mathbf{k},\omega) = \sum_{m=1}^{M} w_m S(\mathbf{x}_m,\omega) e^{i\mathbf{k}\cdot\mathbf{x}_m} \tag{4.44}$$

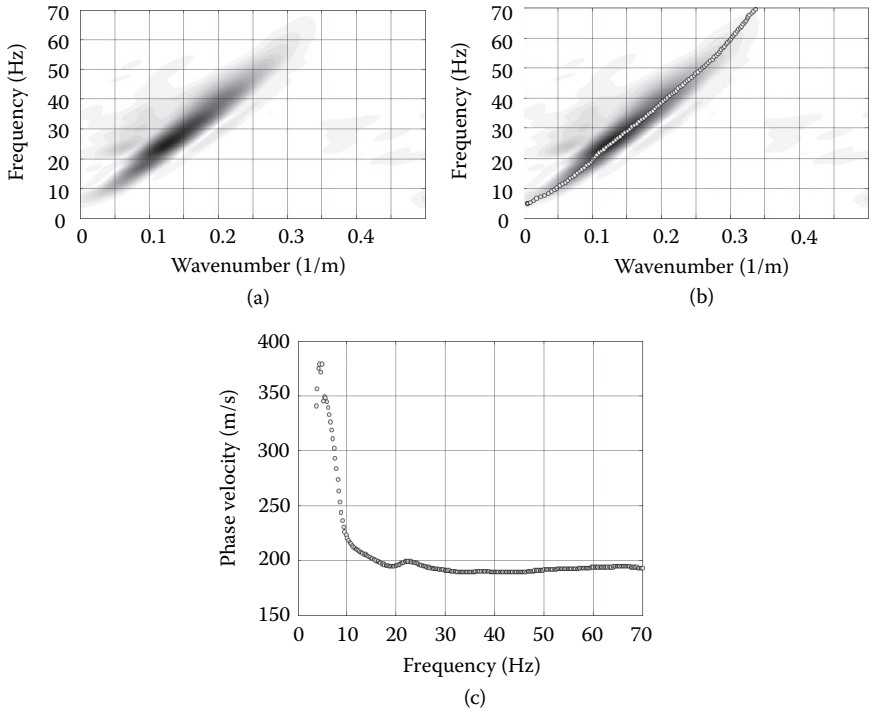

Figure 4.28 Example of dispersion analysis in the frequency domain using the 2D Fourier transform: (a) *f*–*k* spectrum; (b) picking of maxima in the *f*–*k* spectrum; (c) experimental dispersion curve.

where

$$\mathbf{W} = \begin{bmatrix} w_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & w_M \end{bmatrix} \tag{4.45}$$

is a diagonal matrix containing *shading weights* for each receiver. It is convenient to express Equation 4.44 in quadratic form as follows

$$Z(\mathbf{k},\omega) = \mathbf{e}^H\mathbf{WS} \tag{4.46}$$

where H denotes the Hermitian transpose of a vector.

The power in the beamformer output is given by the *steered response power spectrum*

$$P(\mathbf{k},\omega) = ZZ^H \tag{4.47}$$
$$= \mathbf{e}^H\mathbf{W}\mathbf{S}\mathbf{S}^H\mathbf{W}^H\mathbf{e}$$
$$= \mathbf{e}^H\mathbf{W}\mathbf{R}\mathbf{W}^H\mathbf{e}$$

where R is the *spatiospectral correlation matrix*

$$\mathbf{R}(\omega) = \mathbf{S}\mathbf{S}^H = \begin{bmatrix} S_{11}(\omega) & S_{12}(\omega) & \cdots & S_{1M}(\omega) \\ S_{21}(\omega) & S_{22}(\omega) & \cdots & S_{2M}(\omega) \\ \vdots & \vdots & \ddots & \vdots \\ S_{M1}(\omega) & S_{M2}(\omega) & \cdots & S_{MM}(\omega) \end{bmatrix} \tag{4.48}$$

As noted earlier, it is desirable to use ensemble averaging to reduce the variance of the measured cross-power spectra (Bendat and Piersol 2010). Equation 4.47 thus becomes

$$P(\mathbf{k},\omega) = \mathbf{e}^H\mathbf{W}\hat{\mathbf{R}}\mathbf{W}^H\mathbf{e} \tag{4.49}$$

where $\hat{\mathbf{R}}$ denotes a spatiospectral correlation matrix with terms that are the average cross-power spectra between pairs of receivers.

The frequency beamforming technique can be applied to receiver arrays of any shape and is often used to analyze passive data collected with 2D geometries on the ground surface (see Section 7.2.1).

For active-source data with linear arrays, the formulation can be simplified because the source position and the direction of propagation are known; hence, the vector wavenumber $\mathbf{k}$ and the receiver position $\mathbf{x}$ can be simplified to scalar values k_x and x_m. The optimal shading weights are defined according to an approximation of the geometric spreading of Rayleigh waves (Zywicki 1999)

$$w_m = \sqrt{x_m} \tag{4.50}$$

The processing technique proposed by Park et al. (1999) for the multichannel analysis of surface waves (MASW) method is equivalent to a conventional frequency-domain beamforming with shading weights equal to

$$w_m = \frac{1}{|S(x_m,\omega)|} \tag{4.51}$$

so that the normalized amplitude of each receiver is 1.0.

A variety of more advanced signal processing techniques are available that may generally be called adaptive array processing methods (Johnson and Dudgeon 1993). Examples include minimum variance, linear prediction, Pisarenko harmonic decomposition, eigenvector, and multiple signal classification methods. These methods generally provide a higher resolution than the conventional beamforming technique. A detailed presentation of these methods is beyond the scope of this book; interested readers are referred to Johnson and Dudgeon (1993) for a discussion of these advanced methods and to Zywicki (1999) for examples of the application of these methods to engineering surface wave analyses.

4.6.2 Frequency–slowness analysis (MASW)

The use of the frequency–slowness (ω–p) transform for the analysis of dispersive waves was proposed by McMechan and Yedlin (1981). It is based on the slant stack (τ–p) transform or linear Radon transform, one of the basic tools of seismic data processing. The τ–p transform allows the decomposition of a wave field into its plane wave linear components. In this case, the Radon transform is defined as

$$\bar{u}(\tau, p) = \int_{-\infty}^{\infty} u(\tau + px, x)dx \tag{4.52}$$

The transform gives the energy density as a function of τ (the time delay at zero offset) and p (the ray parameter, or slowness). The linear Radon or τ–p transform of a record in time-offset domain stacks the wave field along a straight line of slope p for each value of τ (Figure 4.29a). A straight line in the time-offset domain is described by a constant τ and p; a linear event in the x–t domain with a slope p and intercept τ would map into a single point in the τ–p plane.

On real discrete data, Equation 4.52 is converted to the discrete slant stack transform in which the integral is substituted by a sum

$$\bar{u}(\tau_l, p_k) = \sum_{j=1}^{N} u(\tau_l + p_k x_j, x_j) \tag{4.53}$$

where $\tau_l = l\Delta\tau$ and $p_k = k\Delta p$. For the computation, the minimum and maximum value of p ($p_{\min}$ and $p_{\max}$) and the step Δp must be set a priori; $p_{\min}$ and $p_{\max}$ are easily estimated on the basis of the inspection of the

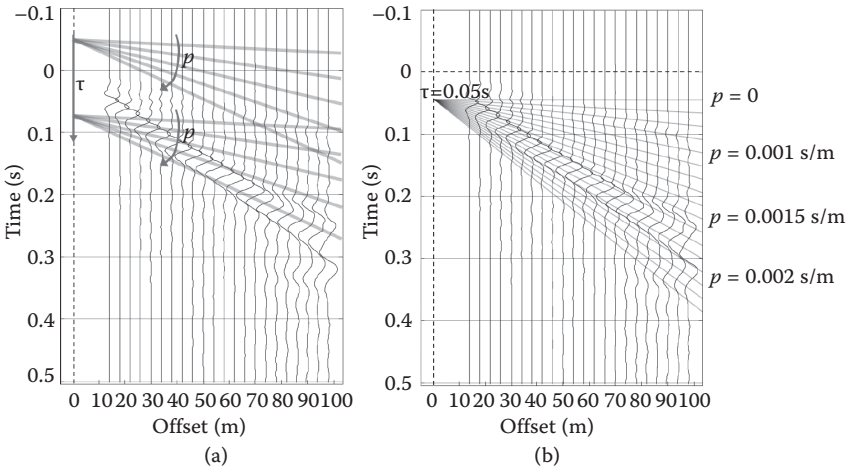

Figure 4.29 (a) Schematic representation of the principle of the τ–p transform, and (b) the data in time offset are summed along straight lines, with intercept τ and slope p.

experimental seismogram; Δp depends on the frequency content of the signal. To avoid aliasing, the following sampling condition has to be satisfied

$$p \leq \frac{1}{2\pi f_{\max} x_{\max}} \tag{4.54}$$

where $f_{\max}$ and $x_{\max}$ are the maximum frequency and the length of the transformed data in the x direction, respectively.

It is important to observe that the τ–p transform can be easily computed using the 2D Fourier transform, that is, the f–k spectrum (Buttkus 2000)

$$U(f,k) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} u(t,x)\, e^{-i2\pi(ft-kx)}\, dx\, dt \tag{4.55}$$

and its inverse

$$u(t,x) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} U(f,k) e^{i2\pi(ft-kx)}\, dk\, df \tag{4.56}$$

If $U(f,k)$ is computed along a straight line $k = fp$ for the fixed slowness value p, then

$$U(f,fp) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} u(t,x) e^{-i2\pi f(t-px)}\, dx\, dt \tag{4.57}$$

Substituting $\tau = t - px$, one gets

$$U(f,fp) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} u(\tau + px, x)e^{-i2\pi f\tau}\, dx\, d\tau \tag{4.58}$$

And remembering that

$$\bar{u}(\tau, p) = \int_{-\infty}^{+\infty} u(\tau + px, x)dx \tag{4.59}$$

we have

$$U(f, fp) = \int_{-\infty}^{\infty} \bar{u}(\tau, p)e^{-i2\pi f\tau}\, d\tau \tag{4.60}$$

which has the inverse Fourier transform

$$\bar{u}(\tau, p) = \int_{-\infty}^{\infty} U(f, fp)e^{-i2\pi f\tau}\, df \tag{4.61}$$

showing finally that the t–p transform can be determined by first transforming the wave field $u(x,t)$ into the f–k domain and then calculating the one-dimensional (1D) inverse Fourier transform along a straight line $k = fp$ for each p value. The details of the algorithm for direct and inverse τp transform can be found in Buttkus (2000).

Figure 4.30 reports a synthetic seismogram consisting of two linear events propagating with different velocities (285 and 133 m/s) and with different initial times. The two events also have different frequencies (16 and 32 Hz, respectively). The two slownesses (0.0035 and 0.0075 s/m) are clearly mapped in the corresponding τ–p transform (Figure 4.30b).

If the τ–p image is Fourier transformed along the time direction, a frequency–slowness image can be produced (Figure 4.31a). The conversion into frequency–phase velocity (Figure 4.31b) is straightforward.

The τ–v and the frequency–phase velocity transform of the reference dataset are reported in Figure 4.32.

In comparison to the analysis in the frequency–wavenumber domain, the localization of peaks in the frequency–slowness domain defined by this technique is directly informative about the shape of the dispersion curve. If the two methods are also equivalent in principle, the one based on the slant stack transform gives a clearer and faster image of the dispersion.

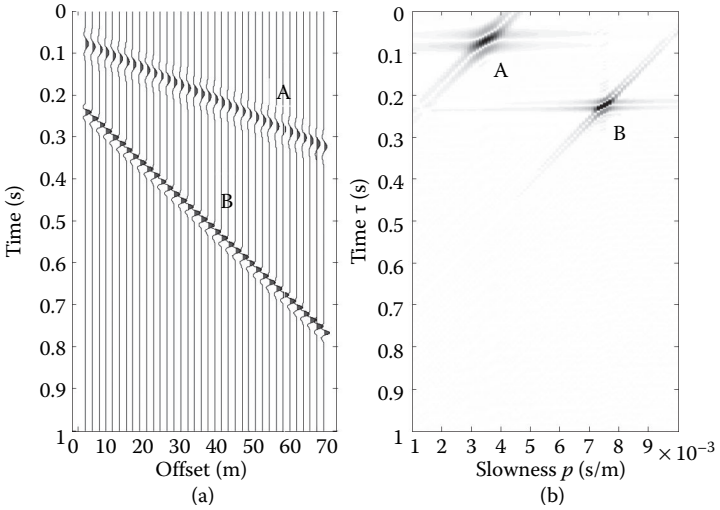

Figure 4.30 Example of τ–p transform of a simple synthetic with two events: (A) velocity = 285 m/s, t_0 = 0.08 s, f = 16 Hz; (B) velocity = 133 m/s, t_0 = 0.22 s, f = 32 Hz): (a) Time-offset data and (b) τ–p transform.

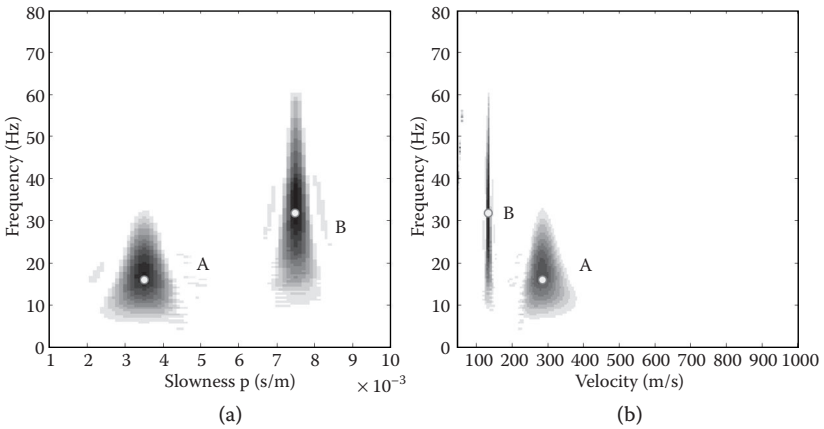

Figure 4.31 (a) Slowness–frequency and (b) velocity–frequency spectra computed from the τ–p transform of Figure 4.30.

4.6.3 Refraction microtremor method

In passive surface wave surveys, the ambient noise (microtremor) is usually recorded with 2D arrays of sensors (Aki 1957; Lacoss et al. 1969; Ohori et al. 2002). The 2D receiver distribution allows the identification of surface wave propagation direction generated by uncontrolled

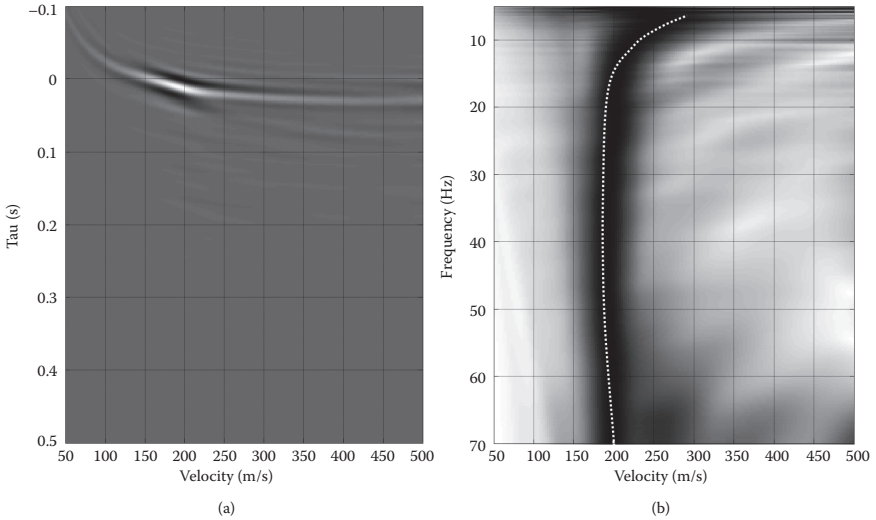

Figure 4.32 Examples of (a) τ–v and (b) frequency–velocity panels for the reference dataset.

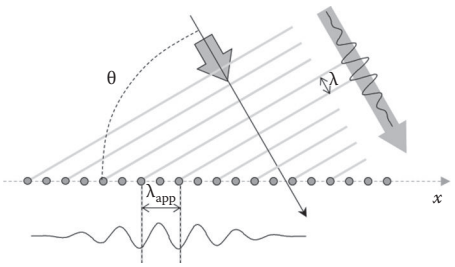

Figure 4.33 When detecting a plane wave with a linear array, an apparent wavelength longer than the true wavelength is observed. Therefore, there is an apparent velocity higher than the true velocity. The apparent in-line wavenumber is smaller than the true wavenumber.

and unknown sources. This approach requires specific acquisition and processing techniques. Louie (2001) proposed an alternative approach to conducting passive measurements using linear arrays and the equipment available for seismic refraction investigations; this approach is known as the refraction microtremor (ReMi) technique.

The obvious limitation is that, in the presence of a single source, the measured apparent velocity depends on the (unknown) angle ϑ between the array direction and the source azimuth (Figure 4.33). For a single plane wave recorded by an array in the direction x, the apparent velocity is equal to the true velocity divided by $cos\vartheta$; thus, it is always greater than

or equal to the actual velocity. The apparent wavenumber k_x (i.e., the component in the array direction of the true wavenumber k) can be written as

$$k_x = k \cdot \cos\vartheta \tag{4.62}$$

Neglecting the spectral leakage due to the finite array length, a single frequency appears as a spike in the wavenumber spectrum.

It is possible to overcome this limitation if the microtremor is an isotropic and diffused wave field, with sources homogenously distributed at all azimuths. In such a situation, considering a single frequency, the k_x–k_y spectrum has a polar symmetry. The spectrum of an ideal diffused wave field with unit wavenumber is represented in Figure 4.34a. A linear array detects not only the in-line energy along the x axis with the positive and negative

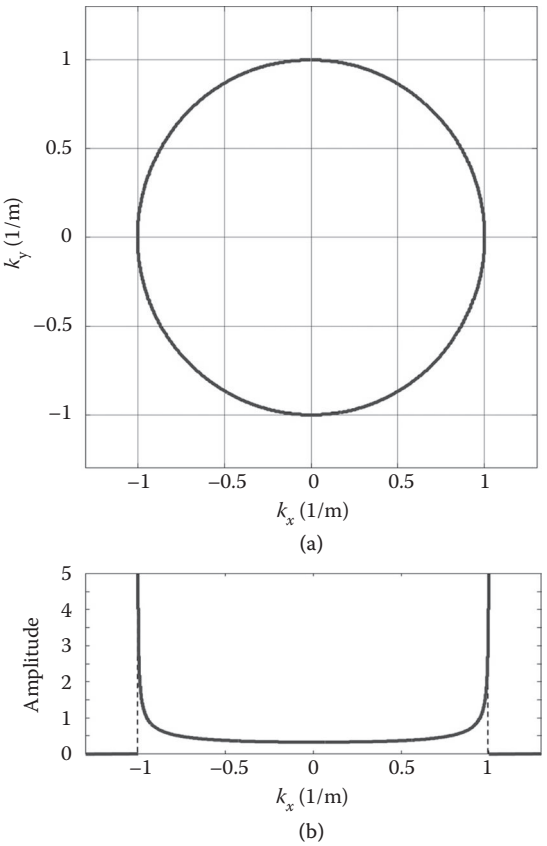

Figure 4.34 (a) The ideal spectrum of a diffused wave field with unit wavenumber in the k_x–k_y domain. (b) The corresponding spectrum observed by a linear array—the spectral maxima are on the true wavenumber.

actual wavenumbers, but it also detects the energy traveling with lower apparent wavenumbers. The true positive and negative wavenumbers are stationary points in the projection onto the array direction, and they result in spectral maxima (Figure 4.34b). Considering an array in the x direction, the spectrum of k_x can be written as (Strobbia and Cassiani 2011)

$$P(E) = \frac{1}{\sqrt{1-\cos^2}} \frac{1}{\pi} \tag{4.63}$$

The spectrum represented in Figure 4.34b does not change with the direction of the array due to the polar symmetry of the wave field.

The finite length of the recording array introduces some complexity in the ReMi spectrum that needs to be considered. Figure 4.35 shows the ReMi spectra for a given frequency for plane waves with uniform azimuthal directions and for two different array lengths (96 and 48 m). Note also that, for any finite length array, the maximum of the 1D spectrum will be at a wavenumber smaller than the true value.

The resulting ReMi spectrum in the f–k domain is plotted in Figure 4.36a for a simple layered medium, assuming a perfectly isotropic diffused wave field. Half of the wave field will have a positive propagation direction and half a negative propagation direction, with respect to the orientation of the x axis. For a homogeneous wave field, the spectrum is symmetric. This property can be used for a check on the assumption of uniform noise distribution in the experimental dataset, which is a fundamental hypothesis for the ReMi method. The symmetry of the spectrum is a necessary condition but not a sufficient condition to prove that the noise wave field is isotropic. A symmetric crossline distribution of sources can produce a symmetric spectrum with a very large overestimation of the actual phase velocity.

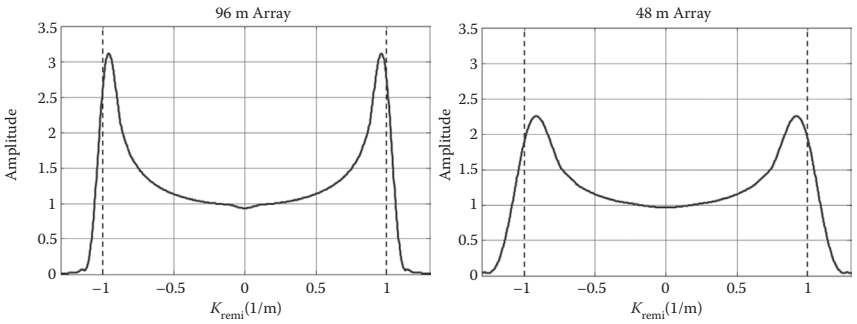

Figure 4.35 The theoretical ReMi wavenumber spectra for two different lengths of the linear array, for a perfectly isotropic noise wave field. The true unit wavenumber is indicated with vertical dashed lines. The maxima are located at values of k smaller than the true wavenumber of the wave field.

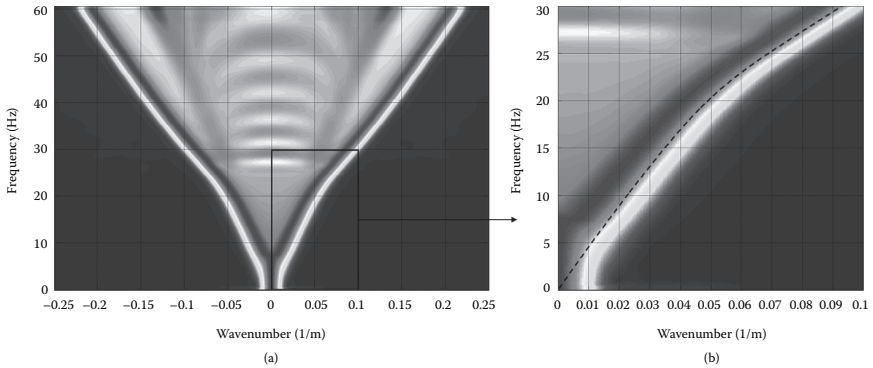

Figure 4.36 (a) Synthetic ReMi spectrum for a perfectly isotropic noise. The spectrum is symmetric; (b) The close-up shows the relationship between the spectral maximum and the true wavenumber (indicated as a black dashed line).

The identification of the true wavenumber from the ReMi spectrum is not straightforward. The true wavenumber is plotted as a dotted line in Figure 4.36b; at each frequency, the maximum of energy is close but not coincident to the true wavenumber. In the original ReMi method (Louie 2001), one way of estimating the wavenumber is by manual picking in a zone between the maximum and the point where the spectral amplitude decreases abruptly. The manual picking is subjective and could introduce large errors in the estimated wavenumber and, consequently, on the experimental phase velocity. To show the difference between the ReMi spectrum and the active data spectrum, with an in-line source, we compare the spectra and the theoretical position of the true wavenumber for a synthetic dataset (Figure 4.37).

Strobbia and Cassiani (2011) showed that, with a uniform distribution of sources, the shape of the ReMi spectrum depends only on the true wavenumber and on the array length. Therefore, the true wavenumber can be identified automatically from the experimental spectrum, without the need for a subjective manual picking. Moreover, the proposed approach can be used to check the assumption of uniform noise distribution in the experimental dataset, which is a requisite to obtain reliable results.

The steps of the method are as follows:

1. The data are segmented into normalized, overlapping subwindows.
2. For each subwindow, the elementary spectrum (either *f–k* or *f–p* or *f–v*) is computed, normalized, and stacked; positive and negative quadrants are plotted.

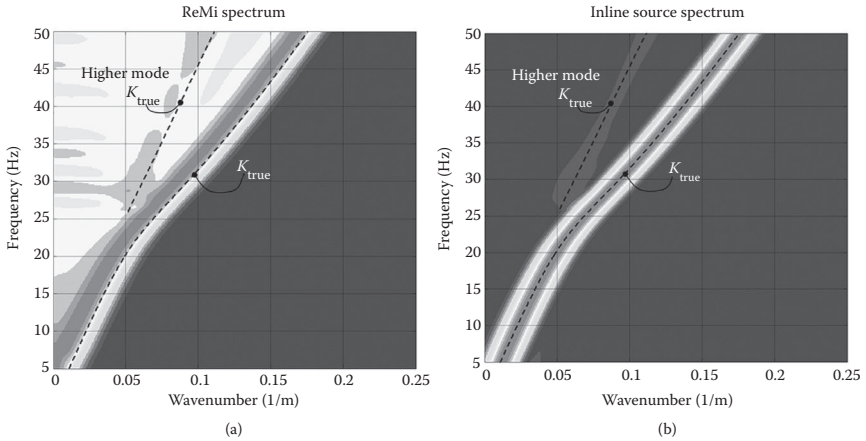

Figure 4.37 Comparison between (a) the ReMi spectrum for a perfectly isotropic noise and (b) the spectrum with an in-line source. The position of the true wavenumber, for the fundamental and first higher modes, is plotted as dashed lines on both spectra. The maximum of the ReMi spectrum is not on the true wavenumber.

3. The frequency range of interest is picked. The spectrum is normalized and split into two parts (negative and positive wavenumbers), which can be independently inverted, fitting the data with the theoretical shape of the spectrum.
4. The two inverted wavenumbers are compared to check the assumption of uniform noise distribution.

An example of ReMi *f–k* spectrum, computed from a 15 min noise record on a 24 channel array, is plotted in Figure 4.38.

The inspection of the spectrum qualitatively shows an acceptable symmetry and suggests the presence of higher modes as secondary local maxima. An example of the fitting of a section of the spectrum is presented in Figure 4.39, considering the frequency of 30 Hz.

The results of the inversion procedure for the whole frequency range are shown in Figure 4.40. Figure 4.40a shows the wavenumbers for the positive and negative quadrants, and Figure 4.40b shows the associated misfit; Figure 4.40c shows the two dispersion curves, together with the curve from an active acquisition obtained on the same array. For this case, the agreement between active and passive data is remarkably good.

Another example is reported in Figure 4.41. In this case, the spectrum is markedly asymmetric, with a sharp variation of the wavenumber at low frequencies. An asymmetric spectrum can indicate the presence of a dominant noise source in one direction.

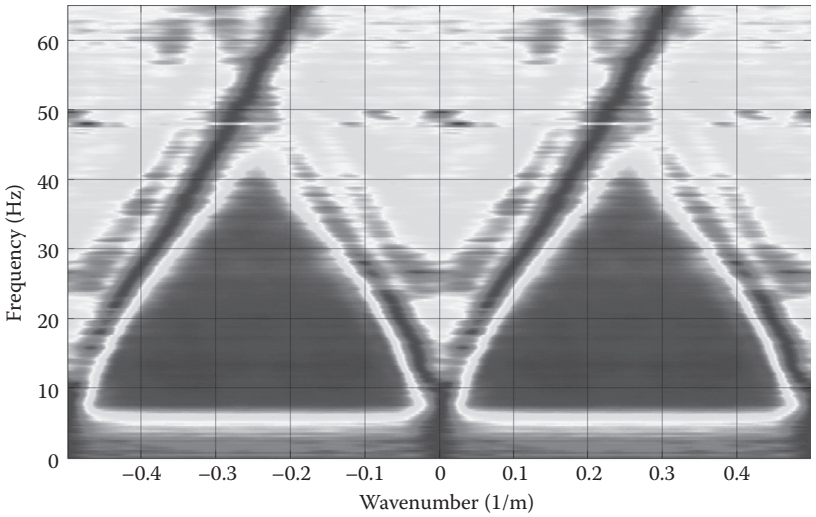

Figure 4.38 Experimental ReMi spectrum for a site where the basic hypothesis of uniform distribution of the seismic sources appears to be verified because of the symmetry of the spectrum.

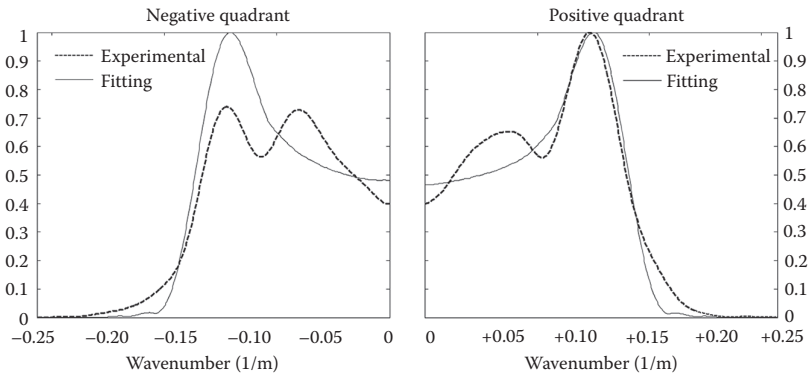

Figure 4.39 The two quadrants of the ReMi *f–k* spectrum (Figure 4.38) are inverted independently to estimate the wavenumber. The fitting of the slice at 30 Hz is shown here.

A very interesting case occurs when the source is *nearly* in-line. Indeed, sometimes the array is deployed purposefully and aligned with some known source of ambient noise in an attempt to mimic active tests. In this case, the spectral shape is similar to those of a single source and the spectral maximum is expected on the true wavenumber. The ReMi spectrum should be picked as an active spectrum in such condition.

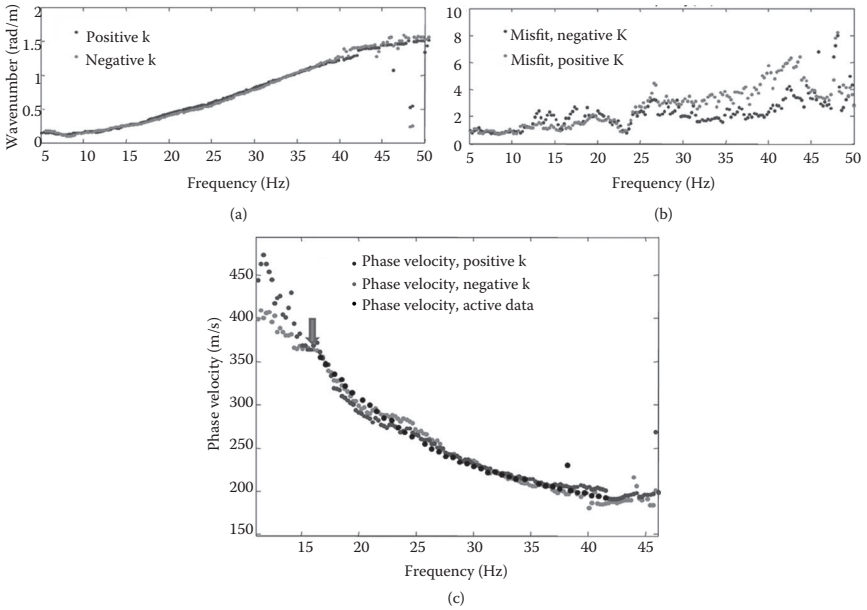

Figure 4.40 Results from the application of the automated ReMi procedure on the dataset of Figure 4.38: (a) estimated wavenumbers for the positive and the negative quadrants; (b) misfit between the experimental spectrum and the theoretical function; (c) experimental dispersion curve from the ReMi procedure and from an active-source test at the same site.

4.7 GROUP VELOCITY ANALYSIS

Analysis of group velocity is used for seismological applications related to the characterization of the Earth's crust on the basis of high-period teleseismic signals. This technique is also applied to microtremors for geological basin characterization, but their limited resolution prevents a diffuse use for near-surface characterization.

The multiple filter method proposed by Dziewonski et al. (1969) for the analysis of earthquake signals is based on the use of band-pass frequency filters. Applying narrow band-pass filters with different center frequencies, different wave groups are separated in the signal (Figure 4.42). These different packets of oscillation can be associated with different modes of propagation, and the peak of each envelope can be used to evaluate the corresponding wave group time delay and hence its group velocity (Figure 4.42b).

The main problem concerning the application of this technique to active-source measurements is related to interference between the modes, which occurs frequently when different modes have similar velocities or when the

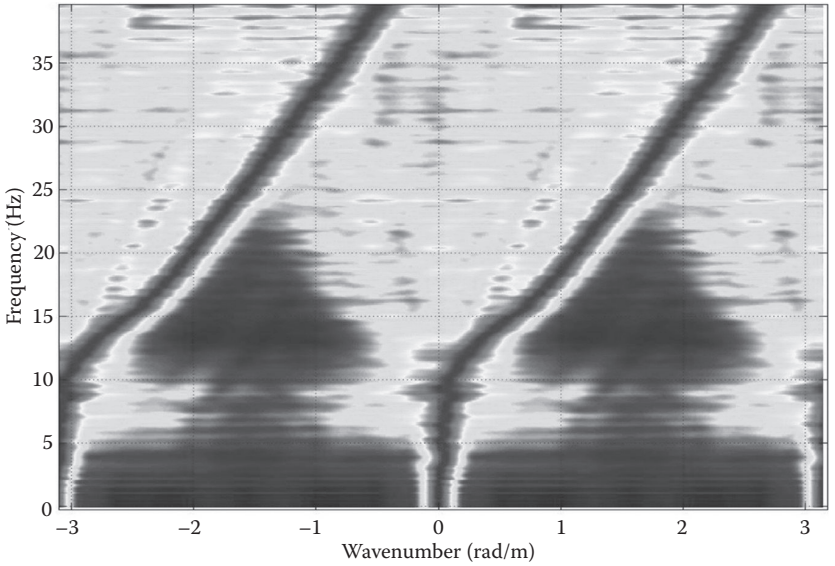

Figure 4.41 Experimental ReMi spectrum for a site where the basic hypothesis of uniform distribution of the seismic sources appears not to be verified because of the lack of symmetry in the spectrum.

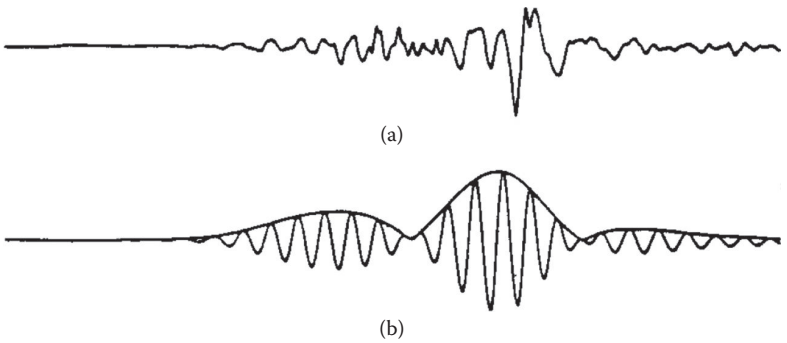

Figure 4.42 Example of narrow band-pass filtering: (a) original earthquake signal; (b) filtered signal. (From Dziewonski, A. et al., *Bull Seismol Soc Am,* 59, 427–444, 1969.)

source is too close to the receivers. Such problems can prevent the method from being effective in resolving the different dispersion curves. Gabriels et al. (1987) reported such difficulties in their attempt to analyze the data collected in a multistation session using the multiple filter technique.

The analysis of group velocity can also be used to separate the contribution of Rayleigh modes before the evaluation of phase velocity (Al-Hunaidi 1994;

Karray and Lefebre 2009). The separation can be obtained with the time-variable filter technique, originally proposed to filter out noise (Pilant and Knopoff 1964) and then applied to the analysis of multimode signals for seismological and geophysical applications (Landisman et al. 1969) to cut off the energy that is not associated with a selected wave group.

4.8 ERRORS AND UNCERTAINTIES IN DISPERSION ANALYSES

Uncertainties in dispersion analysis arise from uncertainties in the experimental data and uncertainties introduced during the processing of the dataset.

Data uncertainties are due mainly to noise in the recorded signals and to geometrical uncertainties related to the location and tilting of the receivers. The influence of several sources of uncertainty has been studied by O'Neill (2003) using numerical simulations. He reports minimal influence from geophone tilt and coupling, whereas positional errors, static shift, and additive Gaussian noise introduce larger uncertainties in the experimental dispersion.

Uncertainties in recorded signals are associated with coherent noise and uncorrelated noise. The latter is externally generated noise (environmental noise) and can be studied via the statistical distribution of the recorded signals, if many repetitions of the test in a given configuration are available. Coherent noise is due to events generated by the seismic source (i.e., near-field effects). An approach for evaluating uncertainties in phase analysis has been provided in Section 4.4.

The estimation of the uncertainty is not trivial when the experimental dispersion curve is determined using transform-based methods. Indeed, difficulties arise in properly quantifying how the data error is propagated through a series of complex data processing steps—from the acquisition of the signals in the time-offset domain (t,x) to the calculation of the dispersion curve.

The uncertainty associated with the experimental dispersion curve in multistation surface wave testing is more easily determined by a direct measurement of the statistical distribution of primitive and derived surface wave data. Even though this approach requires considerably more time and effort to be implemented, it does not involve simplifying assumptions (e.g., small variance of the raw data), and it is exempt from major technical difficulties.

Figure 4.43a reports the ensemble of 12 independent dispersion curves obtained with frequency–wavenumber analysis of each available shot gather for the reference dataset. The experimental dispersion curve can be subdivided into two regions: a high-frequency region with very low values of the coefficient of variation and a low-frequency region with higher values

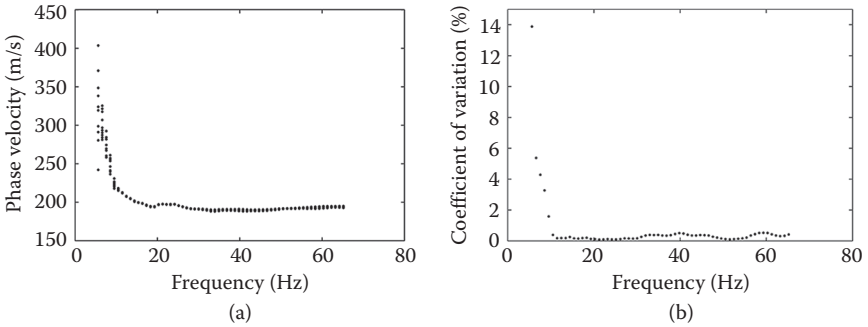

Figure 4.43 Assessment of the uncertainty associated with the experimental dispersion curve: (a) experimental dispersion curves obtained from the frequency–wavenumber analysis of each shot for the same experimental setup; (b) coefficient of variation for the phase velocity at each frequency as obtained from the ensemble of data points in (a).

of the coefficient of variation (Figure 4.43b). These trends are in agreement with other studies on different datasets (Tuomi and Hiltunen 1996; Marosi and Hiltunen 2004a; Lai et al. 2005). Lai et al. (2005) also showed that the assumption of Gaussian distribution is reasonable for the population of phase velocity at each frequency.

The aforementioned uncertainties are limited to repeatability of the test in the same testing setup and with dispersion analysis performed using the same computer code. Larger uncertainties are to be expected because the measurement and processing techniques are varied. In this respect, several comparative analyses at benchmark testing sites provide an insight into the variability that can be expected, even if it is not simple to observe well-defined trends and consistent differences among different processing techniques (Cornou et al. 2007; Boore and Asten 2008; Tran and Hiltunen 2011; Kim et al. 2013). Uncertainties in surface wave analysis are further discussed in Section 6.5.2.